Cleaner Firms, Dirtier Trade:

Carbon Offshoring and the Limits of Production-Based Accounting

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Abstract

Manufacturing in rich countries is getting cleaner—on paper. Recent firm-level studies find little evidence that offshoring to pollution havens drives this clean-up. Linking plant-level fuel use for Swedish manufacturers (2006–2014) to imports weighted by their embodied carbon, I instrument firms' carbon imports with a shift-share of foreign export-supply shocks. A 10 percent rise in carbon imports lowers production-based CO_2 intensity by 6.7 percent. This is not cleaner technique: two thirds of the decline is value added rising, and I find no fuel switching. The pollution-haven margin, sourcing the same products from dirtier origins, is weak on average, as earlier studies find, but significant among firms that bought many distinct products before the shock, which have more room to reallocate. Yet the carbon does not disappear: firms' own transport emissions and the international freight carrying their imports both rise, and the national inventory attributes neither to Swedish manufacturing. The imported goods embody 64 percent more carbon than producing them in Sweden would. At most 2 percent of imported carbon enters through a country where the importing firm owns a plant, so leakage policy aimed at multinationals misses almost all of it, and the EU's carbon border adjustment, which excludes freight, misses the transport leg. Where emissions are counted can matter as much as whether they fall.

JEL: Q54, Q56, F18, F14, F23.

Keywords: carbon offshoring, carbon leakage, freight emissions, carbon footprint, shift-share, Swedish firms

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1 Introduction

Does offshoring make manufacturing cleaner, or does it merely change where the associated emissions are recorded? A growing firm-level literature examines whether firms relocate pollution-intensive production to countries with less stringent environmental regulation, but finds limited evidence of systematic shifts towards more polluting origins (Akerman et al., 2024; Leisner et al., 2026; Dussaux et al., 2023). Yet this evidence addresses only one margin of offshoring. Firms can substitute imported inputs for domestic production without changing the countries from which they source. The broader environmental consequences of this substitution remain an open question.

Manufacturing emissions can fall through two distinct margins: cleaner production techniques or changes in the composition of domestic production. While a substantial literature examines how environmental regulation induces firms to adopt cleaner techniques (Levinson, 2023; Najjar and Cherniwchan, 2021; Brunel, 2017; Agostini et al., 1992; Andersson, 2019; Ponce and Khan, 2021; Marin and Vona, 2021), firms can also reduce domestic emissions by replacing pollution-intensive inputs they previously produced themselves with imports (Baumert et al., 2019; Jakob et al., 2022; Levinson, 2010; Nielsen et al., 2021). This shifts the composition of domestic production without necessarily improving production techniques. The emissions from producing these inputs are then recorded abroad, while those from transporting them fall outside the importing firm's production account. A decline in domestic emissions can therefore reflect a relocation of production rather than a reduction in the emissions associated with firms' production and sourcing.

Sweden provides a particularly useful setting for examining this distinction. It combines stringent domestic carbon pricing with substantial reliance on imported carbon. In 2020, its emissions-weighted carbon price was approximately $71 per tonne of CO_2 (Dolphin and Merkle, 2024), while net carbon imports amounted to 64 percent of its production-based emissions (Friedlingstein et al., 2026).[1] This contrast highlights how stringent domestic climate policy can coexist with substantial carbon embodied in imports.

I examine this possibility using linked data on Swedish manufacturing firms from 2006 to 2014, combining plant-level fuel-use records with information on the carbon embodied in firms' imported inputs. A 10 percent increase in the carbon embodied in intermediate imports reduces firms' stationary-combustion CO_2 emissions per unit of value added by 6.7 percent. This decline in domestic emissions intensity does not necessarily indicate cleaner production or lower

[1] Both series are retrieved from Our World in Data; Figure B.1 places Sweden against the other OECD economies on both margins.

emissions across the firm's production and sourcing activities.

Identifying this effect is challenging because firms choose whether and how much to offshore, and these choices may reflect factors that also affect their emissions. I address this selection problem by instrumenting firms' embodied-carbon imports with a shift-share measure based on foreign export-supply shocks, weighted by firms' pre-period sourcing patterns (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2022). Assessing the broader emissions consequences of this sourcing also requires going beyond the importing firm's production account to measure the carbon embodied in imported inputs and the emissions associated with their transport.

The results reveal how this decline in domestic emissions intensity arises and what it leaves out. I find no detectable change in fuel switching or emissions per unit of energy. Instead, the decline is driven largely by rising value added, rather than by cleaner energy use or production techniques. Decomposing embodied-carbon imports into import volume, product composition and origin dirtiness, I find that the response is concentrated in import volume and product mix. Sourcing the same products from dirtier origins is detectable only among firms with broader pre-existing input portfolios, suggesting that the average absence of origin substitution masks a response among firms with more diversified sourcing relationships. The implications extend beyond domestic production emissions. The imported basket has a median carbon intensity 64 percent higher than producing the same inputs in Sweden would have entailed. Carbon-intensive sourcing is also associated with higher emissions from firms' own transport and international freight, although the latter is not recorded in the importing manufacturer's production account. Finally, at most 2 percent of imported manufacturing carbon enters through countries where the importing firm owns a plant, providing an upper bound on the share potentially associated with intra-firm sourcing.

This paper brings together two aspects of firms' international production decisions that are often examined separately: how sourcing changes domestic emissions intensity, and where the emissions associated with that sourcing are generated. By tracing changes in import volume, product composition and origin, I distinguish shifts in the scale and composition of sourcing from changes in the carbon intensity of suppliers. I then extend the analysis beyond domestic production emissions to include the carbon embodied in imported inputs, firms' own transport and international freight. This broader accounting also allows me to distinguish imports potentially sourced within multinational groups from those purchased from independent suppliers.

These findings have implications for how manufacturing decarbonisation and carbon offshoring are understood. A decline in emissions per unit of value added may reflect changes in the scale and composition of domestic production, rather than cleaner production techniques.

Moreover, production-based accounts omit emissions embodied in imported inputs and their transport, while ownership-based measures can overlook carbon embodied in inputs purchased from independent suppliers. Assessing the environmental consequences of firms' international production decisions therefore requires looking beyond both domestic emissions and foreign ownership to the sourcing relationships through which production is organised.

A firm-level literature examines how importing and offshoring affect the environmental performance of manufacturing firms (Akerman et al., 2024; Dussaux et al., 2023; Li and Zhou, 2017; Choi et al., 2023; Leisner et al., 2026; Barrows and Ollivier, 2021). A related strand investigates whether firms relocate pollution-intensive production to countries with less stringent environmental regulation (Akerman et al., 2024; Dussaux et al., 2023; Leisner et al., 2026). This paper connects these strands by examining how the carbon embodied in imported inputs, rather than their monetary value alone, shapes firms' domestic emissions intensity. I combine origin-specific emissions intensities with firm-level import data to distinguish changes in import volume, product composition, and the environmental characteristics of sourcing origins. I then relate these sourcing margins to the components of domestic emission intensity, distinguishing changes in energy use and the carbon content of energy from changes in value added. This approach makes it possible to examine whether changes in domestic emission intensity associated with importing reflect changes in production techniques, the composition of domestic activity, or the relocation of carbon-intensive production stages.

A separate literature examines the contribution of international trade and environmental regulation to changes in manufacturing emissions (Cherniwchan et al., 2017; Cherniwchan, 2017; Shapiro and Walker, 2018; Copeland et al., 2022; Cristea et al., 2013; Cole et al., 2014; Stavropoulos et al., 2018; Levinson, 2010). Production-based emissions accounts, however, record emissions where production takes place and therefore do not capture the carbon embodied in imported inputs or the emissions associated with transporting them. I extend the firm-level analysis by bringing together plant-level energy use, the carbon embodied in imported intermediate inputs, firms' own transport fuel, and the international freight associated with their imports. Examining these components on a common sample and using the same source of identifying variation provides a consistent way to trace how the emissions associated with firms' sourcing decisions are distributed across domestic production, foreign production, and transport. I also distinguish imports potentially occurring within multinational groups from arm's-length sourcing, allowing me to examine the extent to which foreign ownership captures the cross-border relocation of carbon-intensive production. Taken together, this approach broadens the analysis beyond the emissions recorded at the domestic production site to consider the wider emissions implications

of firms' production and sourcing decisions.

2 Conceptual Framework

A country's emissions inventory draws a boundary around its territory, but a firm's production and sourcing decisions extend beyond it. When a Swedish manufacturer replaces a domestically produced input with an imported one, emissions associated with that input move outside Sweden's production-based account, even if the firm's overall carbon footprint does not fall. The shift also generates emissions from transporting the imported input. Understanding the environmental consequences of offshoring therefore requires distinguishing changes in domestic production emissions from changes in the emissions embodied in imported inputs and their transport.

Let M_{ict} denote firm i's imports from country c in year t, e_{ct} the associated production emissions per unit of imported input, d_c the shipping distance, and $\varphi_{m(c)}$ the emissions per unit of freight associated with the transport mode on that route. The emissions associated with the firm's domestic production and imported inputs can then be decomposed into three components:

$$E_{it}^{\text{ACC}} = E_{it}^{\text{DOM}} + \underbrace{\sum_c M_{ict} d_c \varphi_{m(c)}}_{E_{it}^{\text{FR}}\ \text{(international freight)}} + \underbrace{\sum_c M_{ict} e_{ct}}_{E_{it}^{\text{FGN}}\ \text{(embodied abroad)}} . \tag{1}$$

The first component captures emissions from the firm's domestic production, including domestic transport fuel burned by the firm. The second captures emissions from transporting imported inputs, while the third measures the emissions generated in producing those inputs abroad. The latter is the embodied-carbon import measure used throughout the paper. Unlike domestic production emissions, neither the carbon embodied in imported inputs nor the associated international freight is recorded in the importing firm's production-based emissions.

The difference between these components matters because offshoring can reduce domestic production emissions without reducing the emissions associated with the firm's production and sourcing. When a firm replaces a domestically produced input with an imported one, emissions from domestic production may fall, while emissions embodied in the imported input are generated abroad. Transporting that input also adds freight emissions. Production-based accounting records the domestic reduction but excludes the latter two components. Whether offshoring reduces emissions overall therefore depends on how the emissions avoided in domestic production compare with those embodied in the imported input and its transport. A single leakage measure would obscure these distinct adjustments; examining them separately reveals where the

emissions consequences of sourcing arise.

2.1 The sourcing problem

The emissions components in equation (1) are jointly determined by firms' sourcing decisions. Firms choose not only how much to import, but also which countries to source from. These choices are interdependent: entering a new sourcing country involves a fixed cost, while the attractiveness of each source depends on the firm's overall sourcing portfolio. I build on the multi-country sourcing framework of Antràs et al. (2017) to characterise these decisions and derive the margins along which changes in sourcing affect the emissions associated with firms' production.

Consider a firm i that uses a set of inputs indexed by product $p \in \mathcal{P}_i$. It can obtain these inputs domestically or from foreign suppliers. Establishing a sourcing relationship with country c requires paying a fixed cost $f_c > 0$ in each period, so the firm chooses a set of sourcing countries, Ω_i. Within this set, country c supplies product p at an effective cost $w_{ct}\tau_c/a_{pc}$, where w_{ct} is the input price, τ_c captures trade costs, and a_{pc} represents the productivity of sourcing product p from country c. Productivity draws follow a Fréchet distribution with shape parameter θ.

Under this structure, the firm's sourcing capability is

$$\Theta_{it} = \sum_{c \in \Omega_i} T_c (w_{ct}\tau_c)^{-\theta},$$

and the unit cost of its input bundle is proportional to $\Theta_{it}^{-1/\theta}$. The share of expenditure on product p allocated to country c is

$$s_{ipct} = \frac{T_c (w_{ct}\tau_c)^{-\theta}}{\Theta_{it}}, \qquad c \in \Omega_i. \tag{2}$$

An important implication is that these origin shares do not vary across products: conditional on the firm's sourcing set, the model assigns the same country shares to each product. Product composition is instead governed by the product weights ω_{ipt}. This separation provides a way to distinguish changes in sourcing origins from changes in the composition of imported products. It is also a substantive restriction of the framework, which I return to below.

The carbon embodied in a firm's imported inputs depends on three features of its sourcing: the scale of imports, the composition of the imported product basket, and the countries from which those products are sourced. To separate these margins, I first express embodied-carbon

imports as

$$E_{it}^{\text{FGN}} = M_{it} \sum_{p} \omega_{ipt} \sum_{c \in \Omega_i} s_{ipct} e_{pc},$$

where M_{it} is the value of imports, ω_{ipt} is the expenditure share of product p, s_{ipct} is the share of product p sourced from country c, and e_{pc} is the corresponding product–country carbon intensity. I then benchmark the imported basket against the emissions that would have been generated had the same products been produced in Sweden. Using base-year Swedish intensities, e_{p0}^{SE}, as the reference, and allowing Swedish intensities to vary over time, gives the following decomposition of the firm's emissions footprint:

$$\ln E_{it}^{\text{ACC}} = \underbrace{\ln M_{it}}_{\text{VOLUME}} + \underbrace{\ln\left(\sum_{p} \omega_{ipt} e_{p0}^{\text{SE}}\right)}_{\text{PRODUCT MIX}} + \underbrace{\ln\left(\frac{\sum_p \omega_{ipt} e_{pt}^{\text{SE}}}{\sum_p \omega_{ipt} e_{p0}^{\text{SE}}}\right)}_{\text{DOMESTIC TECHNIQUE}} + \underbrace{\ln\left(\frac{\sum_p \omega_{ipt} \sum_c s_{ipct} e_{pc}}{\sum_p \omega_{ipt} e_{pt}^{\text{SE}}}\right)}_{\text{ORIGIN DIRTINESS}} + \underbrace{\ln\left(1 + \frac{E_{it}^{\text{FR}}}{E_{it}^{\text{FGN}}}\right)}_{\text{FREIGHT}}.$$

The decomposition separates the scale and composition of imports from the emissions intensity of their production and transport. The volume term captures changes in total imports. Product mix captures changes in the composition of the imported basket, evaluated at base-year Swedish intensities. Domestic technique captures changes in the domestic benchmark for producing those products. Origin dirtiness measures how the carbon intensity of actual foreign sourcing differs from that benchmark. The final term adds emissions from international freight. Together, these components account for the emissions associated with domestic production and the firm's imported inputs, including their international transport.

The decomposition is an accounting identity; it does not, by itself, predict how any component responds to a change in foreign supply conditions. Those predictions come from the sourcing model. I therefore turn to the comparative statics of an exogenous change in foreign supply costs.

2.2 Comparative statics

Consider an exogenous decline in supply costs in a subset of foreign origins, such that $d \ln w_{ct} < 0$ for $c \in \mathcal{S}$. The shock makes these origins more attractive to Swedish firms, independently of changes in their own sourcing decisions. Section 5 describes the empirical strategy used to isolate this variation. For the comparative statics, the important feature is that the shock operates across origins rather than across products: it changes the relative attractiveness of sourcing countries and can therefore affect the different components of embodied-carbon imports in distinct ways.

The response of import volume follows from the reduction in sourcing costs. Cheaper foreign inputs increase the firm's sourcing capability, Θ_{it}, and reduce the unit cost of its input bundle. Whether this translates into higher import expenditure also depends on the firm's output re-

sponse. Product composition, by contrast, is not directly affected by an origin-specific shock. Under the model's assumption that product weights, ω_{ipt}, are determined by the firm's technology, the shock does not generate a systematic change in the product-mix margin. Freight emissions may nevertheless rise as imports expand. Their relationship to embodied production emissions depends on how the shock changes the geographical composition of sourcing: if the firm does not systematically shift towards more distant or less distant origins, freight and embodied emissions move proportionately, leaving their relative wedge unchanged. Finally, the domestic-technique benchmark is not directly affected by a foreign supply shock. A response in this component would therefore point to other changes in domestic production conditions or to confounding variation that the empirical design has not absorbed.

The response of origin dirtiness depends more directly on firms' existing sourcing networks. Consider a carbon-intensive origin $c' \notin \Omega_i$ that becomes cheaper. The firm enters this origin only if the expected cost savings exceed the fixed cost $f_{c'}$. Because a sourcing relationship can serve several products, the value of entry depends on the firm's overall input requirements: firms purchasing a broader range of products can spread the fixed cost across more uses. This feature generates complementarity across sourcing decisions, as in Antràs et al. (2017). It also creates heterogeneity in how firms respond to the shock. Firms that already source from the newly cheaper origin can adjust their expenditure shares within their existing sourcing set, changing the origin composition of imports without incurring an additional entry cost. Firms that do not yet source from that origin must first overcome the fixed-cost threshold. For sufficiently small cost reductions, they may not enter at all. The model therefore distinguishes adjustment within existing sourcing relationships from the entry of new origins.

2.3 Predictions

The model yields three predictions that guide the empirical analysis in Section 7.4. They concern the response of product composition, freight emissions, and origin choice to an origin-specific supply shock.

Prediction 1 *An origin-specific supply shock does not directly affect the product composition of the firm's imports. Under the assumption that product weights, ω_{ipt}, are determined by technology rather than sourcing conditions, the product-mix response should be limited and should not vary systematically with the breadth of the firm's sourcing network.*

Prediction 2 *The freight wedge, $\ln(1 + E_{it}^{\mathrm{FR}} / E_{it}^{\mathrm{FGN}})$, remains unchanged if the shock does not systematically alter the geographical composition of sourcing. Freight emissions may nevertheless rise in levels as*

imports expand. A zero response in the freight wedge therefore does not imply that offshoring generates no transport emissions; it implies that freight emissions increase proportionately with the carbon embodied in imports.

Prediction 3 *The response of origin dirtiness depends on firms' ability and incentive to adjust their sourcing networks. Firms already sourcing from an origin can reallocate expenditure towards it without incurring an additional entry cost. For firms not yet sourcing from that origin, entry requires the expected cost savings to exceed the fixed cost. The incentive to enter should be stronger when the firm's input requirements allow that fixed cost to be spread across more products. Consequently, the origin-dirtiness response should be stronger among firms with broader baseline product scope, while the product-mix response should not vary systematically with scope.*

The third prediction is particularly informative because it distinguishes the two composition margins across the same partition of firms. If larger or more import-dependent firms simply respond more strongly to foreign supply shocks, their responses should extend beyond origin dirtiness to other margins, including import volume and potentially product composition. The model instead predicts a specific relationship between baseline product scope and adjustment in sourcing origins, without a corresponding relationship for the product-mix margin.

2.4 Assumptions

The model assumes that, conditional on a firm's sourcing set, its sourcing-country shares are common across products. This restriction allows changes in the composition of imports to be separated into product-mix and origin-dirtiness margins. It rules out product-specific sourcing patterns—for example, a firm sourcing steel from China but chemicals from Germany because of differences specific to those products. The data cannot directly test the common-share restriction. However, the finding that the composition of firms' import baskets at EXIOBASE resolution changes little provides an indirect diagnostic: it suggests that product-composition adjustments are limited in the setting studied here.

The framework is partial equilibrium. Foreign supply costs, w_{ct}, are taken as given, and the model does not allow Swedish output prices or broader market conditions to adjust in response to firms' sourcing decisions. It therefore characterises firms' sourcing responses to foreign cost shocks rather than the general-equilibrium consequences of those shocks, and it does not provide a welfare analysis.

Finally, the fixed cost f_c represents the cost of establishing access to a sourcing origin, not the cost of owning a foreign affiliate. The framework is therefore silent on whether imports

are supplied by affiliated or independent firms. This distinction matters for interpreting the empirical evidence on intra-firm trade: the model concerns the formation and use of sourcing relationships, whereas ownership is a separate organisational choice.

3 Data

Measuring carbon offshoring at the firm level requires three objects that are rarely observed together: a firm's own emissions, the full product-by-origin composition of its imports, and the emission intensity of production in the countries it buys from. I assemble them from Swedish administrative microdata held by Statistics Sweden (SCB), matched to a global multi-region input–output table. This section describes each in turn, then the sample.

3.1 Data sources

SCB collects annual energy use for every manufacturing plant with at least ten employees. Firms report physical quantities by fuel (litres of diesel, MWh of electricity, hundreds of cubic metres of natural gas), not emissions. I convert quantities to CO_2 using fuel-specific combustion factors,

$$E_{it} = \sum_f \alpha_f \, x_{ift}, \tag{3}$$

where x_{ift} is firm i's consumption of fuel f in year t and α_f the CO_2 content of that fuel. The same construction is used by Forslid et al. (2018) and Barrows and Ollivier (2021).[2] Because α_f is a constant, (3) measures *stationary combustion*: the CO_2 released when the firm burns fuel on its own premises. It is not a full scope-1 inventory, since it excludes process emissions from chemical reactions such as calcination in cement or reduction in steel. For Sweden over 2005–2014 this is a modest omission for most of manufacturing but not for all of it, and I note below where it binds. The same energy data give a second, coarser measure of environmental performance, energy supply in gigajoules using calorific values, which is available for every fuel including those whose units I cannot reconcile across the two files. Energy and CO_2 answer different questions. Energy use is local in its incidence and varies sharply with how dirty the fuel mix is, whereas CO_2 is a global pollutant that crosses borders and persists (Parry et al., 2014); comparing them shows whether offshoring moves the local or the global margin. Energy is the primitive and CO_2

[2] SCB uses different fuel names in the energy-use file and the emission-factor file. I map them by hand and take the modal coefficient within each energy identifier, which absorbs variation in spelling. Factors are combustion-only and exclude life-cycle emissions.

the derived object: $\ln(CO_2/VA) = \ln(E/VA) + \ln(CO_2/E)$, an identity I exploit in Section 6.4 to separate how much energy a firm uses from how dirty that energy is.

Holding α_f fixed assumes the carbon content of a fuel does not vary with abatement technology. For Sweden in this period that is reasonable: end-of-pipe carbon capture was not deployed at commercial scale. Second, emissions in levels confound environmental performance with firm size. I therefore work throughout with emission *intensity*, emissions per unit of value added, following Holladay (2016), Rodrigue et al. (2024) and Barrows and Ollivier (2021). No denominator is innocent. Value added and sales respond to prices, which carry no environmental content, while physical output requires units to be comparable across products and time and so covers fewer firms.

I match the energy data to three further SCB registers. The Structural Business Statistics (FEK) provide employment, industry (NACE Rev. 2), sales, value added, and balance-sheet items. The International Trade in Goods (ITG) register records the value and quantity of every import and export line by CN8 product and partner country. The Production of Commodities (IVP) register gives production quantities and values at the 8-digit product level. I keep NACE 10–33 (manufacturing) and the years 2005–2014, using 2005 as the pre-sample year from which instrument shares are taken.

Offshoring is not the same as importing. A firm that buys finished goods for resale is not relocating production, and counting all imports conflates production inputs with final goods and with goods never made domestically (El-Sahli et al., 2022). I therefore count only intermediate inputs, classified using the fifth revision of the Broad Economic Categories (BEC codes 121, 220, 420, 521, 530). Because BEC revisions reclassify products over time, I harmonise codes across revisions using the concordance algorithm of Pierce and Schott (2012) as extended by Van Beveren et al. (2012).

I keep NACE 10–33 not only because imports originate mostly there but because dirty industries are systematically more trade-exposed, facing lower tariffs and non-tariff barriers (Shapiro, 2021).

3.2 The carbon content of imports

The object of interest is not the value of these imports but the carbon embodied in them. Between 22 and 35 percent of global emissions are embodied somewhere in the value chain of traded goods (Copeland et al., 2022), and that share has been rising (Eaton et al., 2016), so the intensive margin of the toxic content of each import line is what matters, not its value (Dussaux et al.,

2023). I construct

$$CarbImp_{it} = \sum_{p} \sum_{c} M_{ipct}\, EI_{pct}, \tag{4}$$

where M_{ipct} is firm i's import of product p from country c and EI_{pct} is the emission intensity of producing p in c.

The intensities EI_{pct} come from EXIOBASE, a multi-region input–output table covering 44 countries and 5 rest-of-world regions at 200 products. I use *embodied* intensities, $\mathbf{S}(\mathbf{I} - \mathbf{A})^{-1}$, which count emissions generated anywhere in the upstream chain of a product and not only at the final stage. Intensities are industry-by-country, not firm-by-product, because producer-level intensities for source countries do not exist; this is the standard proxy in the literature (Levinson, 2010; Li and Zhou, 2017), and it is a source of measurement error I return to in Section 5. Appendix P gives the construction in full, including the product concordance, the treatment of zeros and missing years, and the collapse of trade partners into EXIOBASE's rest-of-world regions. Import lines are matched to EXIOBASE products through an HS-to-EXIOBASE crosswalk and to EXIOBASE regions through a country bridge; intensities are expressed in kg CO_2 per euro, and SEK import values are converted at annual Eurostat reference rates.

The earlier literature uses WIOD or EORA and keys intensity on the importing firm's own industry. That construction gives every one of a firm's import lines the same intensity regardless of what the firm actually bought, so the product composition of imports never enters the measure at all. The source matters for the same reason: in EORA the median country-year contains eight industries, some spanning ten Swedish industry divisions, and country explains 63% of the variance in log intensity against 23% for industry. EXIOBASE also separates direct from embodied intensity, and the two correlate only 0.70 in logs.

The difference is large enough to decide whether an instrumental-variables design is feasible at all. Instrumented with the shift-share that Section 5 builds, the EORA firm-keyed measure yields a first-stage F of 0.96, against 6.68 for the underlying import volume on the same sample: the emission weighting, not the instrument, was destroying the first stage. Moving to product-keyed EXIOBASE embodied intensity restores it. Source and key were varied one at a time, so the difference is attributable to the change that produced it.

I also compute, for each firm-year, the counterfactual intensity of the same import basket had it been produced in Sweden, using Swedish EXIOBASE intensities for the same products. The ratio of the two is the object that makes offshoring an environmental question instead of an accounting one: in the sample described below the median imported basket is 1.64 times as emission-intensive as its Swedish counterpart, and the mean 2.18.

To ask whether offshoring tracks policy arbitrage I also use the OECD Environmental Policy Stringency index, a composite of roughly fifteen market and non-market instruments, available 2004–2014. I classify an origin as lax if its index lies below Sweden's in the same year. Appendix C describes the index and the classification.

Emissions embodied in production are not the whole footprint of an import. Moving goods emits as well, and a firm that switches from a domestic to a distant supplier adds transport emissions that no production-side measure captures. I assign each import line a distance from Sweden to the origin country and a modal emission factor, and aggregate to firm-year freight CO_2; Appendix Q gives the construction. Freight is small relative to embodied production emissions (a median of 2.4% of imported carbon, a mean of 5.0%), but it enters the decomposition in Section 6.4 as one of the five margins, and its sign matters because it works against the firm, not for it. Mode is inferred from geography, never observed, and import weights are missing for part of the sample, so the series is a lower bound: where I cannot assign a mode I assign sea, the least carbon-intensive of the four. Appendix Q gives the construction and plots the level and the share over the window.

3.3 Sample and summary statistics

Table 1 traces the sample from the universe of Swedish firm-years to the estimating sample. Two restrictions do nearly all the work. Requiring manufacturing takes 5.1 million firm-years to 366,354. Requiring reported energy use, the binding constraint since SCB surveys only plants with ten or more employees, takes that to 29,738, or 8.1% of manufacturing firm-years. Requiring positive value added and intermediate imports leaves 18,535, and a non-missing instrument 15,763. Matching freight gives the estimating sample of 15,749 firm-years and 3,181 firms.

The sample is 4.3% of manufacturing firm-years and 83.4% of the CO_2 reported by manufacturing. That gap is the point: emissions in manufacturing are concentrated in large plants, which are precisely the plants SCB surveys and precisely the plants that import. A sample covering a small share of firms and most of the emissions is the normal case in this literature, and it means the estimates speak to the bulk of the manufacturing carbon footprint even though they do not speak to the median Swedish manufacturer.

Table A.3 in the appendix compares firms in the estimating sample with manufacturing firms outside it. The estimating sample is larger and more emission-intensive by construction, because it is importing manufacturers. That is a statement about scope, and every specification below is identified within firm.

Table A.2 reports the estimating sample in levels. The median firm employs 47 people, sells

Table 1: From the firm universe to the estimating sample

Restriction	Firm-years	Firms	% mfg	Mt CO_2	% mfg CO_2
All firm-years, 2005-2014	5,126,653	956,583	1399.4	114.68	116.2
manufacturing	366,354	56,863	100.0	98.69	100.0
+ emissions reported	29,738	5,138	8.1	98.69	100.0
+ value added positive	29,566	5,124	8.1	98.26	99.6
+ intermediate imports (BEC)	18,535	3,640	5.1	96.96	98.2
+ both EXIOBASE intensities	18,535	3,640	5.1	96.96	98.2
+ instrument non-missing	15,763	3,183	4.3	82.28	83.4
+ freight matched *(estimating sample)*	15,749	3,181	4.3	82.28	83.4

Notes. Restrictions are cumulative. Shares are expressed relative to manufacturing, not to the full firm universe, so the first row exceeds 100%. CO_2 is stationary combustion computed from reported fuel use; it is defined only for firms that report energy, so the CO_2 column is constant across the first two restrictions by construction and is a share of *reported* manufacturing emissions from the third row on. The estimating sample is 4.3% of manufacturing firm-years and 83.4% of reported manufacturing CO_2.

96 MSEK, and emits 76 tonnes of CO_2 from stationary combustion; the mean firm emits 5,225 tonnes. The distance between those two numbers, a factor of 68, is the concentration described above and the reason every specification below includes firm fixed effects and clusters at the firm.

Three features of the import data shape what follows. First, sourcing is narrow: the median firm imports 4 distinct EXIOBASE products from 5 distinct origins in its first sample year, with means of 5.1 and 7.0. Sourcing sets this small are what make the fixed-cost structure of Section 2 the right framework, and they are what make product-keyed intensity informative: with four products, which four matters. Second, imported baskets are dirtier than their Swedish counterparts, at a median intensity of 0.61 against 0.37 kg CO_2 per euro. Third, baskets change: the median firm turns over 4% of its basket relative to its own first year, but the mean is 26% and the 90th percentile is complete replacement. Most firms source stably, and a minority reallocate wholesale. That heterogeneity, not the average, is what the instrument exploits.

4 Carbon-Intensity Gaps and Sourcing Patterns

Section 3 described the sample. This section establishes the three facts the rest of the paper rests on: that the carbon intensity of production differs sharply across the countries Swedish firms buy from, that the gap holds industry by industry and is no artefact of national averages, and that it survives all the way down to the individual firm's own import basket.

The firm-level fact is the one that matters for identification, so I state it first. For the median firm in the estimating sample, the goods it imports carry **1.64 times** the carbon that producing the same basket in Sweden would have released, at a median import intensity of 0.61 against

a Swedish counterfactual of 0.37 kg CO_2 per euro (Table A.2). The mean ratio is 2.18, so the distribution is skewed and a minority of firms buy baskets several times dirtier than the domestic alternative. Reallocating a given basket toward these origins therefore raises embodied carbon even when the firm buys exactly the same products in exactly the same quantities. That is the margin the instrument is built to move.

Where the gap comes from. The policy-stringency split is descriptive and the identification does not rest on it. Splitting the instrument by origin group leaves no first stage, so the paper cannot say whether the gap reflects environmental policy, the energy mix, or the sectoral composition of what each country exports. Appendix C plots emission intensity by income group and by policy stringency: Sweden sits far below both comparison groups throughout the period and the gap does not close. What the argument needs is weaker and better identified. The gap is large, it is sectoral, and it is priced into the basket a firm buys, so reallocating a given basket toward these origins raises embodied carbon with no change in what is bought.

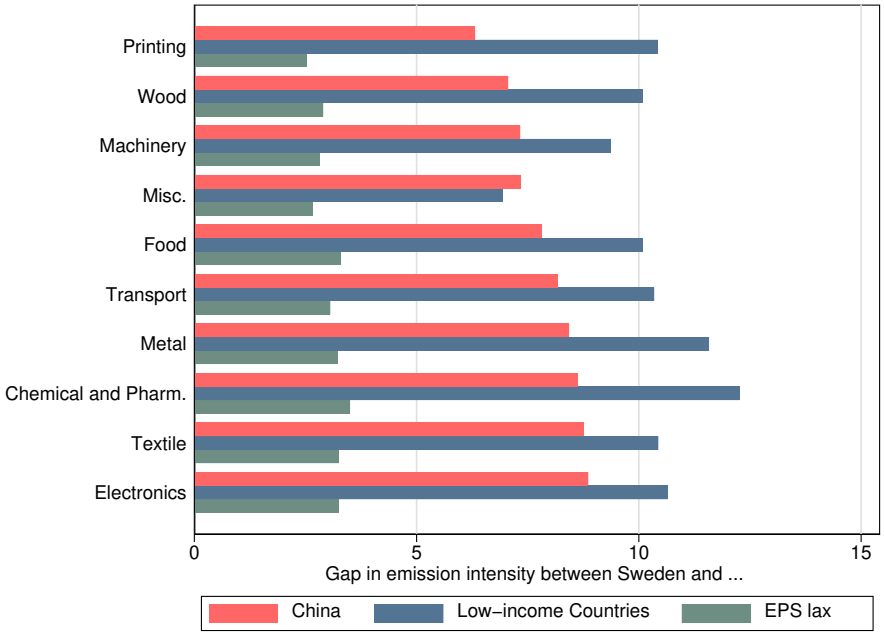
The gap holds industry by industry, and Figure 1 shows its size industry by industry. Sweden is roughly eight times more carbon efficient than China, ten times more efficient than the average low-income country, and four times more efficient than countries whose environmental policy is laxer than its own. The largest gaps are in metals, transport and chemicals, the energy-intensive industries; but textiles and electronics, which are not energy-intensive, show large gaps too. That pattern is itself informative: if the gaps tracked environmental policy, they would not appear where energy is a small share of costs. What the figure establishes is that reallocating a given basket toward these origins raises its embodied carbon substantially, whatever the reason the gaps exist.

Scale, composition and technique. Before turning to firms, it is worth asking what the aggregate record can and cannot establish. Following Choi et al. (2023), Copeland et al. (2022) and Ustyuzhanina (2022), I decompose Swedish manufacturing CO_2 into the scale of the sector, the mix of industries within it, and the emission intensity of each industry. Appendix D gives the construction.

Figure 2 reports it, indexing aggregate manufacturing CO_2 to 1 in 2006. Scale works against the decline, since output grew over the period, and composition works against it too: the sector moved toward more pollution-intensive goods, which is what Ustyuzhanina (2022) also report for Swedish manufacturing over 2007–2017. Technique had to offset both.

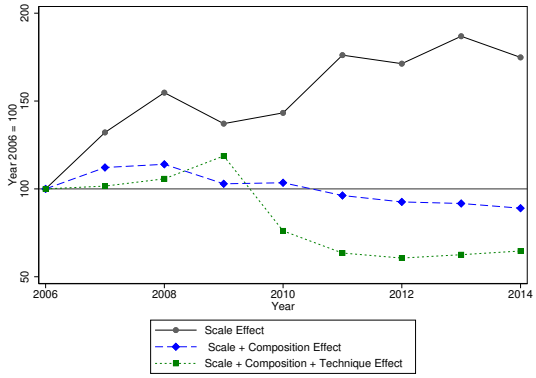
The decomposition cannot go further than that, and the reason is why this paper exists. The composition term counts any shift in the industry mix, whatever produced it, so a shift towards cleaner domestic production is observationally identical here to one that moves dirty production

Figure 1: Sectorial Gap in emission intensities



abroad. No decomposition built on emissions released inside Sweden can separate them. The firm-level analysis below can.

Figure 2: Decomposition of aggregate manufacturing emissions



Notes. Aggregate Swedish manufacturing CO_2 from equation (10), indexed to 1 in 2006. The black line is scale, the gap to the blue line is composition, and the gap to the green line is technique. The composition term counts any shift in the industry mix, whatever produced it.

Overall, despite a substantial increase in manufacturing output measured in monetary value, air pollution emissions from Swedish manufacturing firms fell by 29%. In comparison with most other countries, and despite its limited size, Sweden has a long tradition of strict environmental policies. In 1991 a carbon tax was introduced as a complement to the existing system of energy taxes (Johansson, 2000; Jonsson et al., 2020). The system has changed several times, but a com-

mon feature is higher rates for large carbon-emitting industries, and the country now levies the highest carbon tax rate in the world, at SEK 1,409 (US$129.89) per metric tonne of CO_2.[3] These regulations have historically mattered for reducing emissions from both stationary and mobile sources. Although they may have moved some firms towards cleaner technologies, the measured reduction also reflects the composition of net trade flows across industries. The effect of the carbon tax on the energy and resource efficiency of Swedish manufacturing can therefore be limited, once offshoring becomes an alternative to abating at home.

Concentration of Carbon offshoring. To give a rich background on the historical trends and changes in offshoring, I show in Figure R.1 the concentration of carbon offshoring within 2-digit industries between 1997 and 2014. I use these periods because the trade data go back to 1997, and I have enough sample years to create pre-sample product shares for my empirical purpose. The share of all firms engaged in offshoring, weighted by labour shares and sectors' labour weights, declined from 27.5% in 1997 to 23.0% in 2005. However, this share gradually increased from 2006 (the beginning of my sample period) and reached 27.0% by 2014. Offshoring tends to be concentrated among a select few firms. On average, the five largest offshoring firms within each industry account for 76.6% of the total offshoring volume. Moreover, the concentration becomes even more pronounced when considering local markets. Specifically, the two largest offshoring firms within each industry and county contribute, on average, 87.7% of the total offshoring volume. These descriptive findings highlight the dynamic relationship between offshoring activities, firm emissions, and the importance of properly controlling for firm characteristics in the regression analysis, as firms may self-select into offshoring. This also suggests that after 2005, the offshoring trends have been stable within sectors which is crucial for the construction of my instrument.

5 Empirical Strategy

The relationship between embodied-carbon imports and domestic emissions is potentially endogenous. Firms may import more because they are expanding production, while changes in production and sourcing decisions may occur simultaneously. Consequently, the correlation between embodied-carbon imports and domestic emission intensity need not reflect the effect of sourcing on domestic production. Identifying this effect requires variation in imports that is independent of firms' contemporaneous production decisions.

I address this endogeneity problem using a shift-share instrument that combines firms' pre-

[3] See the World Bank carbon pricing dashboard.

sample product–country import shares with subsequent changes in foreign export supply. The instrument exploits differences in firms' initial sourcing patterns and their resulting exposure to export-supply movements in foreign markets. Its identifying assumption is that, conditional on the fixed effects, these movements affect Swedish firms' domestic emission intensity only through their effect on embodied-carbon imports.

5.1 Estimand and baseline specification

I estimate the relationship between embodied carbon in intermediate imports and firms' domestic emission intensity using the following specification:

$$\ln\left(\frac{E_{it}}{VA_{it}}\right) = \alpha \ln CarbImp_{it} + \theta_i + \theta_{kt} + \theta_{mt} + \varepsilon_{it}, \tag{5}$$

where E_{it}/VA_{it} is firm i's stationary-combustion CO_2 emissions per unit of value added in year t, and $CarbImp_{it}$ is the carbon embodied in its intermediate imports, constructed as described in Section 3. The specification includes firm fixed effects θ_i, industry-by-year fixed effects θ_{kt}, and municipality-by-year fixed effects θ_{mt}. Standard errors are clustered by firm.

The coefficient α captures the elasticity of domestic emission intensity with respect to embodied-carbon imports. Since emission intensity is measured relative to value added, a decline may reflect lower emissions, higher value added, or both. Distinguishing between these possibilities is important for interpreting the estimates, since a reduction in production-based emission intensity does not necessarily imply a reduction in firms' absolute emissions.

Firm fixed effects absorb time-invariant differences across firms, including persistent differences in production technology and sourcing patterns. Industry-by-year fixed effects account for shocks common to firms within the same industry and year, while municipality-by-year fixed effects absorb local shocks that may affect firms' production and energy use. Together, these fixed effects ensure that the estimated relationship is not driven by persistent firm characteristics or shocks common to firms operating in the same industry or municipality.

I progressively introduce these fixed effects across specifications and use the specification including all three sets as my preferred specification. This choice is motivated by the placebo evidence in Section 5.3: without municipality-by-year fixed effects, the instrument predicts changes in domestic production conditions that it should not affect directly. The comparison across specifications therefore provides evidence on the importance of accounting for local shocks when identifying the effect of embodied-carbon imports.

OLS may not identify α because firms' sourcing and production decisions are jointly deter-

mined. Changes in demand or production conditions may affect both imports and domestic emission intensity, while firms may simultaneously adjust their sourcing and abatement decisions. In addition, embodied-carbon imports are measured using country–industry emission intensities rather than firm-specific production technologies. This introduces measurement error into the regressor. I therefore use foreign export-supply movements, interacted with firms' pre-sample sourcing patterns, to instrument embodied-carbon imports.

5.2 Instrument construction

I construct the instrument from firms' pre-sample sourcing patterns and subsequent changes in foreign export supply. For each firm, I measure initial exposure using the shares of intermediate imports sourced from each product–country pair in 2005. I then interact these shares with the corresponding country's exports of each product to the rest of the world, excluding Sweden and its close trading neighbours (Denmark, Norway, Finland, and Germany). Finally, I weight each product–country pair by its base-period EXIOBASE emission intensity, EI_{pc}. Formally, the instrument is:

$$Shock_{it} = \sum_{p} \sum_{c} \bar{\omega}_{ipc} \, \overline{EI}_{pc} \, WX_{pct}, \qquad Z_{it} = \ln(Shock_{it}). \tag{6}$$

The shares fix what a firm was exposed to before the sample opens: which products it bought, and from where. Movements in world export supply then change the conditions facing those particular relationships. Two firms in the same industry and the same municipality, facing identical local conditions, are therefore exposed to different shocks, because in 2005 they were buying different things from different places.

The emission-intensity weights align the instrument with the endogenous variable. An instrument based only on export movements and initial import shares would capture changes in import exposure, irrespective of the carbon content of the goods concerned. This distinction matters because an increase in exports from a relatively clean origin need not generate the same increase in embodied carbon as an equivalent increase from a more carbon-intensive origin. Weighting by base-period emission intensities therefore allows the instrument to capture changes in predicted embodied-carbon imports rather than import exposure alone.

I use base-period emission intensities throughout, so that contemporaneous changes in source-country carbon intensity do not enter the instrument through its weights. This construction does not, however, rule out a separate concern: foreign export movements may coincide with product-level shocks that affect Swedish firms' production directly, independently of their imports. I examine this possibility in Section 5.3. I exclude exports destined for Sweden and

its neighbouring countries to reduce the influence of Swedish demand on the export-supply measure. An increase in Swedish demand could otherwise raise exports to these destinations while simultaneously affecting the production and emissions of Swedish firms. Excluding these markets reduces this direct connection, although it does not by itself ensure that movements in exports to other destinations are exogenous to Swedish firms' production conditions.

The identification strategy follows the shock-based approach to shift-share instruments discussed by Goldsmith-Pinkham et al. (2020) and Borusyak et al. (2022). Rather than requiring firms' initial import shares to be exogenous, this approach requires subsequent export-supply shocks to be conditionally unrelated to shocks to Swedish firms' domestic emission intensity. This distinction is important in the present setting: firms' initial sourcing patterns are endogenous choices and may reflect persistent differences in production technologies, input requirements and access to foreign markets. The identifying assumption concerns the subsequent movements in foreign exports, not the initial allocation of import shares. Conditional on the fixed effects, these movements must be unrelated to other changes in Swedish firms' emission intensity that are correlated with their initial sourcing patterns.

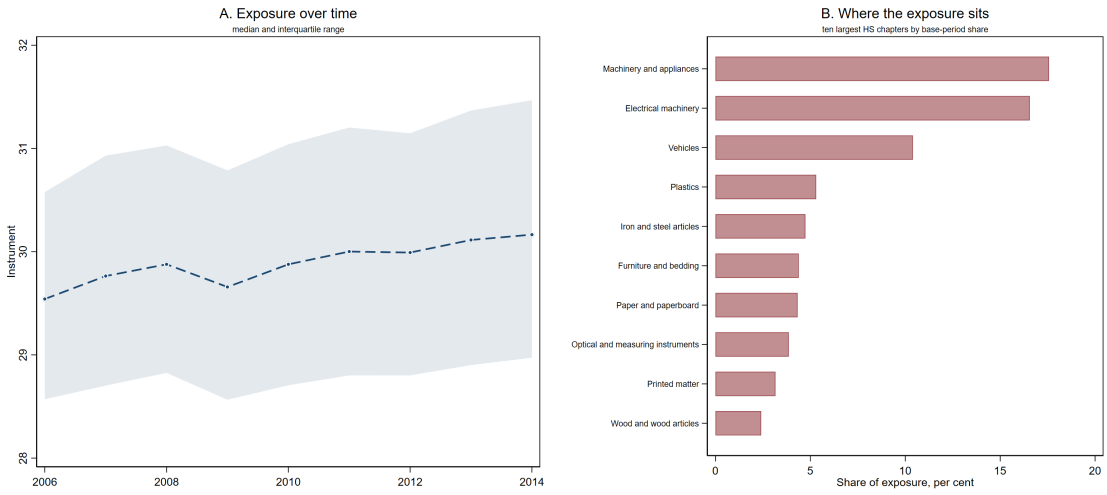
Figure 3 illustrates the distribution of the instrument across firms and over time. Exposure is concentrated in a relatively small number of product groups: machinery, electrical machinery and vehicles account for 44.5 percent of base-period exposure. The first stage in Table 2 is therefore estimated using substantial variation net of persistent firm differences and common industry and local shocks.

5.3 Identification and threats to validity

The identifying assumption is that, conditional on the fixed effects, foreign export-supply movements interacted with firms' pre-sample sourcing shares are unrelated to other shocks affecting their domestic emission intensity. A potential violation arises if export-supply movements coincide with product-specific changes in production technology or input costs that also affect Swedish firms directly. Because firms within the same industry use different combinations of intermediate inputs, industry-by-year fixed effects need not absorb these shocks.

Consider steel. A technological improvement in steelmaking could increase exports from major steel-producing countries while also lowering the production costs or changing the emission intensity of Swedish firms that use steel as an input. Firms with greater pre-sample exposure to these exporters could therefore experience changes in domestic emissions even if their imports did not change. The instrument would then capture a direct effect of foreign production conditions rather than the effect of embodied-carbon imports.

Figure 3: The instrument



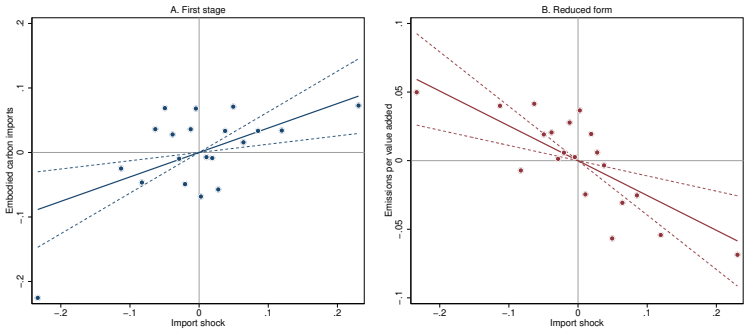
Notes. Panel A plots the median firm-level instrument by year with the interquartile range; the median and not the mean, because a shift-share mean is dragged by the chapters Panel B identifies. Panel B gives the ten Harmonized System chapters carrying the largest share of base-period exposure, where a chapter's share is the sum of firms' pre-sample product shares falling in it; machinery, electrical machinery and vehicles together carry 44.5 percent. *Sources:* UN Comtrade and EXIOBASE, as in Section 3.

I assess this concern using a placebo outcome constructed from the emission intensity of Swedish production of the products in each firm's import basket. This measure captures changes in domestic production conditions for the products to which firms are exposed, but is not mechanically affected by an individual firm's import decisions. If the instrument predicts this outcome, it would indicate that foreign export-supply movements are correlated with changes in Swedish production conditions beyond their effect on embodied-carbon imports.

The placebo results reveal this concern in specifications without municipality-by-year fixed effects. With firm and industry-by-year fixed effects alone, the instrument predicts the placebo outcome. Once municipality-by-year fixed effects are included, the relationship is no longer statistically distinguishable from zero. I therefore use the specification with all three sets of fixed effects as my preferred specification. The placebo does not rule out every possible direct effect of foreign export-supply movements, but it shows that the specifications without municipality-by-year effects retain a relationship with domestic production conditions that is not present in the preferred specification.

The estimated coefficient is also unchanged when I control for firms' effective carbon price. This suggests that the main result is not driven by differences in firms' exposure to carbon pricing. Since the effective carbon price may itself respond to firms' production and fuel choices, I interpret this specification as a robustness check rather than an alternative identification strategy.

Figure 4: First stage and reduced form



Notes. Binned scatters after absorbing firm, sector-year and municipality-year effects, so each slope is the corresponding coefficient. **Left:** the shock against embodied carbon imports, the first stage. **Right:** the shock against emission intensity, the reduced form. The 2SLS estimate in Table 2 is the ratio of the two slopes.

Figure 4 illustrates the first-stage and reduced-form relationships for the preferred specification. Both variables in each panel are residualised with respect to firm, sector–year and municipality–year fixed effects. The left panel shows a positive relationship between the instrument and embodied-carbon imports, while the right panel shows a negative relationship between the instrument and domestic emission intensity.

Taken together, the placebo evidence motivates the inclusion of municipality-by-year fixed effects and supports the preferred specification. The first-stage and reduced-form relationships show how the instrument is associated with embodied-carbon imports and domestic emission intensity in this specification. The causal interpretation nevertheless rests on the conditional exclusion restriction that, given the fixed effects, foreign export-supply movements affect domestic emission intensity only through embodied-carbon imports.

6 The Measured Clean-up

6.1 Embodied-carbon imports and domestic emission intensity

Table 2 reports the relationship between embodied-carbon imports and firms' domestic emission intensity. Column (1) presents the OLS estimate. Columns (2)–(4) instrument embodied-carbon imports using the shift-share design described in Section 5. Column (3), which includes firm, sector–year and municipality–year fixed effects, is the preferred specification. Columns (5) and (6) provide complementary estimates using the value of imported intermediate inputs rather than their embodied carbon.

Under the identifying assumptions set out in Section 5, the preferred estimate is an elasticity of -0.670: a 10 percent increase in the carbon embodied in a firm's intermediate imports is

Table 2: Offshoring and firm-level emission intensity

Instrument	–	Carbon-weighted				Unweighted
	OLS	IV	IV	IV	IV	IV
	(1)	(2)	(3)	(4)	(5)	(6)
Carbon import	-0.025***	-0.628**	-0.670**	-0.657**		
	(0.006)	(0.307)	(0.281)	(0.280)		
Carbon tax rate				-1.137*		
				(0.658)		
Intermediate Import					-0.785**	-0.343**
					(0.350)	(0.175)
Control	No	No	No	Yes	No	No
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Municipal-Year FE	No	No	Yes	Yes	Yes	Yes
Observations	14,825	14,825	14,825	14,825	14,825	14,821
KP		12.20	15.97	15.72	14.87	25.15

Notes. The sample is Swedish manufacturing firms importing intermediate inputs, 2006–2014. The dependent variable in all columns is log CO_2 over value added, where CO_2 is the firm's own stationary combustion. Column (1) reports OLS and columns (2)–(6) report 2SLS. Column (3) is the preferred specification; column (4) adds the firm's effective carbon tax rate. The regressor is log embodied carbon imports in columns (1)–(4) and the log value of imported intermediate inputs in columns (5) and (6), which are estimated on the column (3) sample. The instrument is a shift-share of world export supply on pre-sample product shares. Carbon-weighted means the product shares are weighted by the emission intensity of the exporting industry–country; column (6) leaves the same shift-share unweighted, which is the specification estimated by Akerman et al. (2024). The reported *F*-statistic is the first-stage Kleibergen–Paap *F*-statistic. Standard errors clustered by firm are in parentheses. *** 1%, ** 5%, * 10%.

associated with a 6.7 percent reduction in its stationary-combustion CO_2 emissions per unit of value added. The outcome measures emissions from fuel burned at the firm's own site. The estimate therefore captures a decline in domestic, production-based emission intensity, not necessarily a reduction in the firm's absolute emissions or in the global emissions embodied in its production and imported inputs.[4]

The estimate is similar across the IV specifications. With firm and sector–year fixed effects, the coefficient is -0.628; adding municipality–year fixed effects gives -0.670. It remains close to this value, at -0.657, when I additionally control for the firm's effective carbon tax rate. The similarity across specifications indicates that the estimate is not particularly sensitive to the inclusion of municipality–year fixed effects or the additional tax-rate control. Since the effective tax rate may itself respond to firms' production and fuel choices, I treat column (4) as a sensitivity check rather than as the preferred specification.

[4]The estimate is obtained from Swedish manufacturing firms importing intermediate inputs during the estimation period. It therefore describes the response among firms represented in this sample, rather than the effect of inducing manufacturing firms with no imported inputs to begin importing.

The OLS estimate is much closer to zero, at -0.025. This difference is consistent with the endogeneity and measurement problems discussed in Section 5, although it does not identify which of these accounts for the gap between OLS and IV.[5]

Columns (5) and (6) provide a benchmark using the value of imported intermediate inputs rather than their embodied carbon. Column (5) retains the carbon-weighted instrument, while column (6) uses the unweighted shift-share instrument employed by Akerman et al. (2024). The latter provides the closest comparison with their specification. I estimate an elasticity of -0.343, compared with -0.56 in Akerman et al. (2024). The samples are not nested, so I do not test equality between the estimates; each estimate lies within the other's confidence interval.

Comparing columns (3) and (5) isolates the role of measuring imported inputs by their embodied carbon rather than their value. The two specifications use the same instrument, outcome, sample and fixed effects, and therefore share the same reduced form. Their different IV coefficients arise from differences in the first stage: the instrument moves import value by approximately six-sevenths as much as it moves embodied-carbon imports. Consequently, the estimated elasticity using import value is somewhat larger in absolute terms (-0.785 against -0.670).

Import expenditure captures the value of foreign inputs, but not the carbon embodied in them. This distinction matters for assessing the aggregate implications of firms' sourcing decisions. An increase in import expenditure need not correspond to a proportional increase in embodied carbon, since the carbon intensity of the imported product mix can also change. The aggregate analysis therefore uses embodied-carbon imports to assess how changes in sourcing contributed to the evolution of manufacturing emission intensity.

6.2 Firm-level responses and the aggregate emissions decline

The firm-level response does not necessarily translate into an aggregate contribution to the manufacturing emissions decline. Whether it does depends on how embodied-carbon imports changed over the period and how production and value added evolved across firms.

Between 2006 and 2014, Swedish manufacturing emission intensity fell by 94.4 log points. The aggregate decomposition attributes 86.4 log points of this decline to changes within continuing firms, with the remaining 8.0 log points accounted for by reallocation and entry. Applying the estimated elasticity to firms' observed changes in embodied-carbon imports yields an attributed contribution of 18.6 log points in the opposite direction: because embodied-carbon imports fell over this period, the estimated elasticity implies an increase in domestic emission intensity. Thus,

[5] Weak-instrument-robust Anderson–Rubin confidence sets are reported in Appendix K. These provide inference that remains valid under weak identification, conditional on the validity of the instrument's exclusion restriction.

this margin of offshoring did not drive the aggregate manufacturing clean-up; its implied contribution went against it.

Import values increased over the period, but the carbon embodied in those imports followed a different trajectory. The distinction matters for the aggregate calculation: growth in expenditure on imported inputs does not necessarily imply growth in the carbon content of those inputs. The contribution of sourcing to the evolution of domestic emission intensity therefore depends on changes in the composition and carbon intensity of the imported basket, not simply on import growth.

The aggregate calculation applies the estimated firm-level elasticity to observed changes in embodied-carbon imports. Carrying this relationship to the aggregate requires that the estimated effects survive aggregation: in particular, that one firm's offshoring does not displace emissions from other firms and that input and output prices do not adjust. The calculation is therefore an accounting exercise under these assumptions, rather than a separate counterfactual estimate of national emissions. Appendix D presents the aggregate decomposition and examines the implications of these assumptions, including how large displacement would have to be to overturn the result.

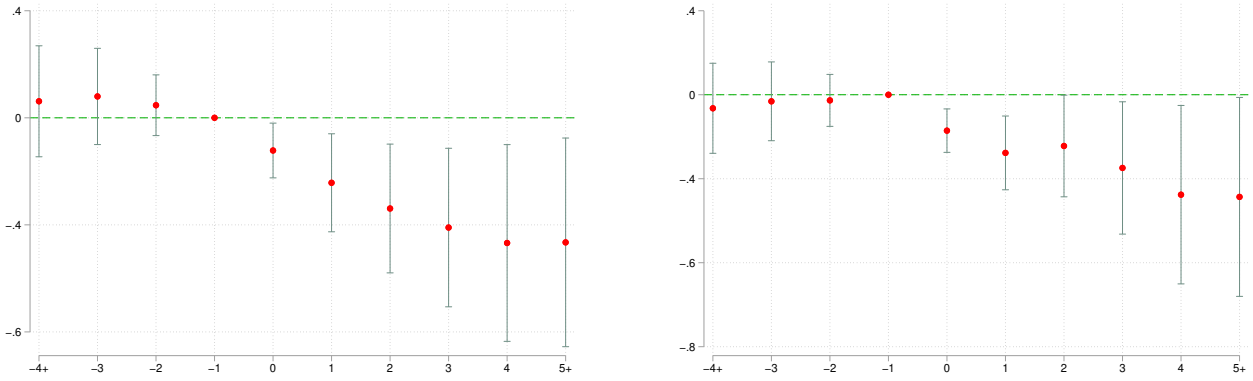
6.3 Robustness

Two concerns motivate the robustness analysis. The first is whether the estimated effect is sensitive to the sample and specification choices underlying the preferred estimate. The second is whether the instrument captures a pre-existing decline in firms' emission intensity rather than variation induced by foreign export-supply shocks. The first concern is about the stability of the estimate; the second bears directly on its causal interpretation.

Figure L.1 examines the sensitivity of the preferred estimate to a set of alternative choices: excluding the financial crisis years, adding policy controls, trimming the instrument at the first and ninety-ninth percentiles, winsorising the outcome, and clustering by firm and sector–year. Across these specifications, the estimate retains its negative sign and remains close to the preferred estimate. Appendix L provides a more systematic assessment, varying the fixed effects, clustering, baseline-scale control and denominator one at a time.

The more fundamental concern is whether the instrument captures a decline in emission intensity that was already under way. If firms more exposed to foreign export-supply shocks were already on a different emissions trajectory before those shocks occurred, the timing of the instrumented response would be difficult to interpret causally. Figure 5 traces the instrumented response to the largest shock over event time, normalising the year before the shock to zero. The

Figure 5: The clean-up is dynamic, not a one-year blip



A. CO_2 per unit value added

B. Energy per unit value added

Notes. IV event-study coefficients β_m, $m = -4, \ldots, 5$, with $\beta_{-1} = 0$; 95% confidence intervals clustered by firm; balanced 2006–2014. Leads sit on zero and the decline begins at the shock and keeps growing. Carbon and energy intensity trace the same path, so the response is a change in how the firm runs, not a one-off accounting shift. Intervals widen at long horizons because fewer firms are observed four and five years out.

pre-shock coefficients are statistically indistinguishable from zero, while the decline begins at the time of the shock and grows over the following four to five years. Confidence intervals widen at longer horizons as fewer firms remain observed.

Appendix J provides a complementary test using the reduced form. Rather than instrumenting firms' imports, it constructs event-time indicators from the timing of each firm's largest export-supply shock and regresses emission intensity directly on these indicators. The pre-shock coefficients are jointly indistinguishable from zero, and the post-shock pattern resembles that in Figure 5. Because this exercise does not divide by the first-stage coefficient, it provides complementary evidence that the observed timing is not driven by the dynamics of the first stage.

The two panels of 5 show similar dynamic responses in CO_2 emissions and energy use per unit of value added. This parallel movement matters for interpretation: the decline in emission intensity is accompanied by a decline in physical energy use, rather than reflecting only a change in the measurement or accounting of emissions. The figure therefore points to a change in firms' energy use relative to output, although it does not yet establish what drives that change.

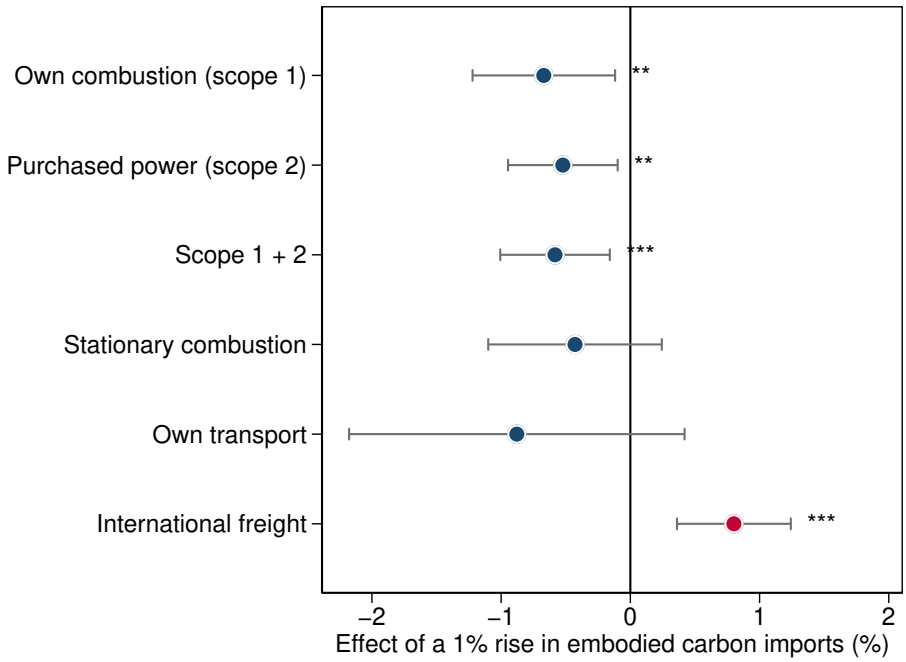
6.4 Which emissions move

Emission intensity aggregates several emissions components and scales them by firm value added. The headline elasticity reported in Table 2 therefore does not reveal which components account for the response. This section examines how the effect is distributed across the firm's emissions account, distinguishing stationary combustion, own transport and purchased energy. Throughout, I retain the instrument, fixed effects and sample used in column (3) of Table 2,

changing only the dependent variable. Section 6.5 then decomposes the decline in emission intensity into changes in energy use and in the carbon content of energy, while Section 7 examines the emissions associated with the firm's imports and their transport.

Figure 6 re-estimates the specification in Table 2, column (3), using each component of the firm's emissions account as the dependent variable, scaled by value added. The outcomes are related by accounting identities: scope 1 and scope 2 sum to scope 1+2, while stationary combustion and own transport partition scope 1. The estimates should therefore be read as a decomposition of the response across components, not as independent effects that can be added together. The figure shows which parts of the firm's emissions account respond to the increase in embodied-carbon imports.

Figure 6: Where the emission effect goes



Notes. Each row is a separate 2SLS regression of that component per unit of value added on log embodied carbon imports, instrumented as in Table 2. Bars are 95% confidence intervals. Components are nested, not additive: scope 1 is own combustion, the sum of stationary and transport fuel; scope 2 is purchased electricity and heat. **International freight is the only component that rises, and the only one outside the national inventory.** Emissions from moving goods to Sweden are attributed to bunkers and to the exporting territory, so a firm that offshores looks cleaner in the production-based account by construction. Firm, sector-year and municipality-year effects; standard errors clustered by firm. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Own combustion falls by – percent for a one percent increase in embodied-carbon imports, while purchased power falls by – percent. The response therefore extends beyond fuel burned directly on the firm's premises. Purchased electricity is a distinct margin: its decline cannot be

attributed simply to a change in the fuels used in on-site boilers. The fact that both components respond is consistent with a broader reduction in the energy used in domestic production, rather than a change confined to the firm's on-site fuel mix. It does not, on its own, establish which production activities have changed or where they have gone.

Within scope 1, the distinction between stationary combustion and own transport helps locate the response. Stationary combustion covers fuel burned on the firm's premises and is therefore more directly connected to on-site production. Own transport captures fuel used in the firm's transport activities. The estimates for these two components indicate whether the response is concentrated in production-related energy use or extends to the firm's own transport activities. Their interpretation, however, depends on the precision of the component estimates.

Own transport deserves separate attention. Research on trade and the environment has focused largely on emissions from production, while the environmental costs of moving goods have received comparatively less attention (Copeland et al., 2022). The firm's own transport fuel captures only part of this margin. It excludes the international freight emissions generated by shipping the intermediate goods it imports, which are not recorded in the importing firm's own transport account. The measure is therefore a lower bound on the transport emissions associated with the firm's sourcing decisions. I examine international freight separately below.

6.5 Energy use and the fuel switching

A firm can reduce its emissions per unit of value added either by using less energy per unit of value added or by using energy with a lower carbon content. These two margins are related by the accounting identity

$$\ln(\mathrm{CO}_2/VA) = \ln(E/VA) + \ln(\mathrm{CO}_2/E).$$

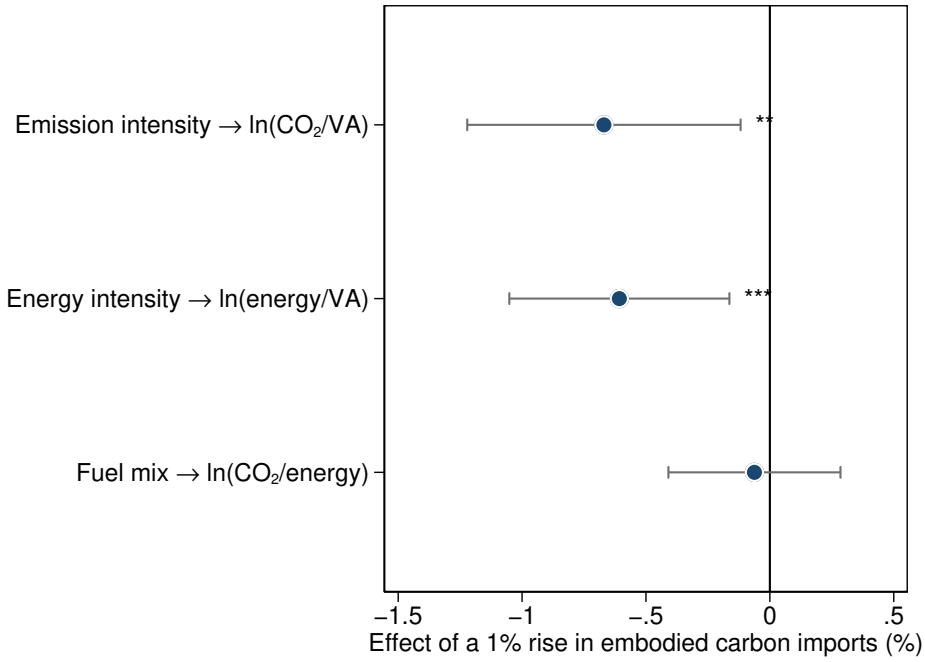
Panel A of Figure 7 decomposes the headline elasticity along these two margins. Of the estimated elasticity of -0.67, the contribution from energy use per unit of value added is -0.60, while that from the carbon content of energy is -0.06. The latter is not statistically significant. The decomposition therefore points to lower energy use per unit of value added, rather than a detectable change in the carbon content of the energy firms use.[6]

An increase in the carbon embodied in imported inputs does not mechanically reduce the energy a Swedish plant needs to produce its existing output. Two explanations could account

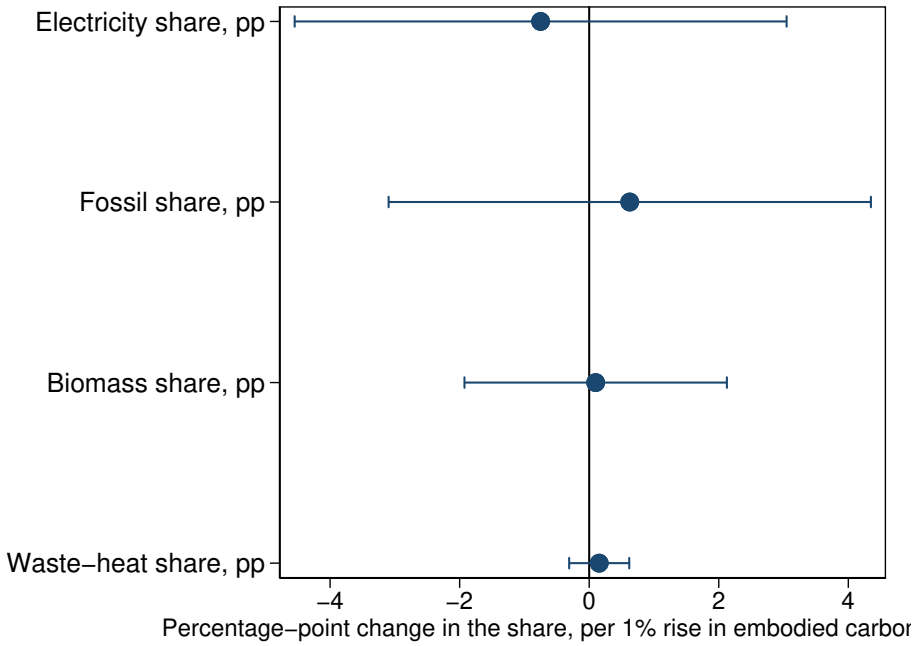
[6] Appendix M repeats the decomposition using the underlying energy and CO_2 quantities rather than their intensities. The distinction matters because the survey records energy use, while CO_2 emissions are calculated by applying fuel-specific emission factors to that use. The two components therefore add exactly by construction. Appendix N reports the component decomposition under each of the three fixed-effects specifications, showing whether the pattern is sensitive to the inclusion of sector–year and municipality–year fixed effects.

Figure 7: Less energy, or cleaner energy?

A. The headline elasticity split into its two terms



B. The fuel shares themselves



Notes. Panel A splits the headline elasticity using $\ln(\mathrm{CO}_2/VA) = \ln(E/VA) + \ln(\mathrm{CO}_2/E)$, an identity, so the lower two rows sum to the top one. The decline is energy per unit of output, not a change in what the firm burns. Because the carbon-content term enters that identity as a residual, panel B tests it directly: shares of total GJ on the left-hand side, in percentage points, at a Kleibergen–Paap F of 22.39. No share moves. **These firms already draw 77 percent of their energy as electricity, so little switching margin remains.** Unlike $\ln(\mathrm{CO}_2/E)$, panel B does not condition on positive combustion CO_2, so firms that switched completely remain in the sample.

for the decline in energy use per unit of value added. Firms may produce the same activities more efficiently, using less energy for a given level and composition of output. This is a technique effect. Alternatively, they may change the composition of domestic production by sourcing inputs from abroad and relocating energy-intensive stages of production. This is a composition effect. The distinction matters for interpreting the decline in domestic emission intensity: a technique effect would represent an improvement in how production is carried out, whereas a composition effect would reflect a change in where production takes place, without necessarily reducing the emissions embodied in the goods consumed.

The decomposition provides little evidence of a change in the carbon content of energy, but that term is a residual in the accounting identity. I therefore test for changes in the energy mix directly, using the shares of total gigajoules consumed by fuel type. Panel B of Figure 7 reports these estimates. The first stage is strong by the standards of the paper, with a Kleibergen–Paap F-statistic of 22.4, but none of the estimated changes is statistically distinguishable from zero. The electricity share falls by 0.75 percentage points, the fossil-fuel share rises by 0.63 percentage points, and emissions per gigajoule rise by 1.36 percentage points. The standard errors are close to two percentage points. Thus, the estimates provide no detectable evidence that firms respond by changing their energy mix; they do not establish that every form of within-plant efficiency improvement is absent.

Appendix E reports the corresponding exercise for abatement investment. The instrument has little power in the survey subsample, so this exercise is uninformative about whether firms change their abatement investment. The fuel-share specification does not condition on positive combustion CO_2, meaning that firms which switch entirely away from fossil fuel remain in the sample. A complete switch would therefore be represented in the estimates.

The limited response in fuel shares may reflect the energy mix of the firms in the sample. Before the shock, these firms already obtain 77 percent of their energy from electricity, leaving a relatively small share in direct fossil-fuel use that could be displaced through fuel switching. This feature of Swedish manufacturing limits the scope for extrapolating the result to settings where industrial energy use is more heavily reliant on fossil fuels. In those settings, firms could have greater scope to respond to the same shock by changing their fuel mix.

The shock also changes the scale of domestic activity. Value added rises by 0.348 and sales by 0.305, both statistically significant at the one percent level, with first-stage F-statistics above 13. These estimates come from a specification that differs from the preferred specification in Table 2, column (3). Re-estimating the outcomes with the preferred set of fixed effects gives a value-added elasticity of 0.463. I use this specification for the decomposition because it is consistent with the

fixed effects underlying the headline estimate. Appendix F reports the corresponding estimates for all outcomes. The estimates for emissions in levels and measured total factor productivity are imprecise, with first-stage F-statistics of 8.6 and 9.7, respectively. They do not establish that either outcome is unaffected by the shock.

Overall, the results are more consistent with a change in the composition of domestic production than with a response through fuel switching. The estimated carbon-content margin is statistically insignificant, the energy mix does not detectably change, and value added rises while energy use per unit of value added falls. This pattern is consistent with firms expanding less energy-intensive activities while sourcing more energy-intensive inputs from abroad. It does not, however, directly establish that particular production stages have moved overseas, nor does it rule out efficiency improvements that are not captured by the fuel-use measures. The evidence identifies a change in the energy intensity and scale of domestic activity; the interpretation in terms of production relocation remains a mechanism consistent with that evidence, rather than a directly observed outcome.

7 The Limits of Production-Based Accounting

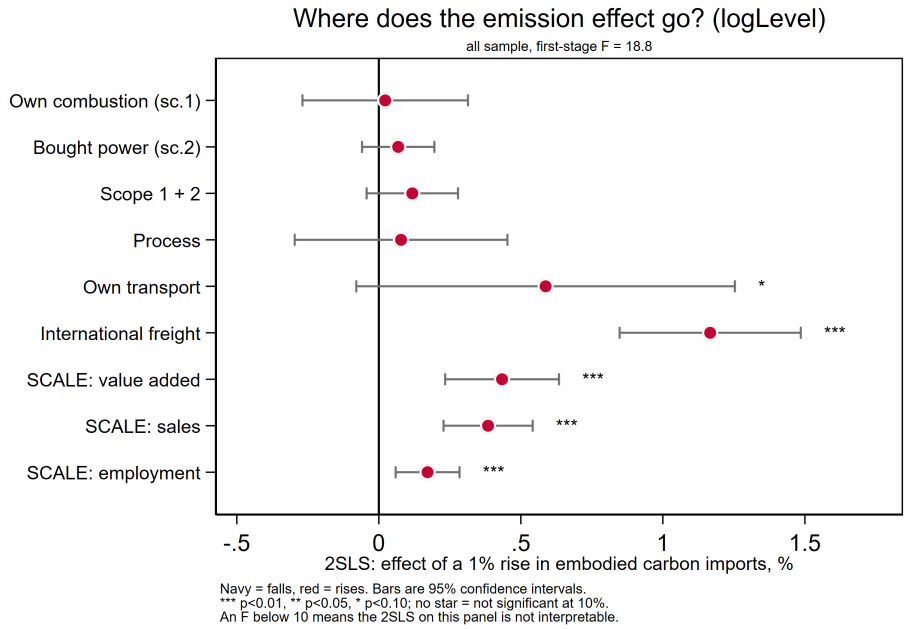
National greenhouse-gas inventories are organised around territorial emissions: they count emissions released within a country's borders. Sweden's national inventory follows this convention, as do the production-based emissions measures used in the firm's carbon-tax and emissions-trading frameworks and in much of the firm-level literature on manufacturing clean-up. This accounting boundary leaves some trade-related emissions outside the manufacturing accounts. It excludes fuel used by firms in their own transport activities from the emissions of manufacturing production, and international shipping emissions are recorded separately from the importing firm's production emissions. These boundaries matter for interpreting the results: the shock increases both firms' own transport emissions and the international freight emissions associated with their imports.

7.1 Output or emissions?

A decline in emission intensity need not reflect a reduction in emissions: it can also arise because value added increases while emissions remain unchanged. To distinguish these possibilities, Figure 8 reports the components of the firm's emissions account in levels, without scaling them by value added. This makes it possible to assess whether the estimated decline in intensity is accompanied by a corresponding change in the emissions themselves.

The estimates reveal a contrast between emissions recorded in the firm's production account and those associated with moving its goods. The estimates for on-site combustion, purchased power, scope 1 and 2 emissions combined, and process emissions are statistically imprecise; they do not establish that these components are unaffected by the shock. By contrast, the estimates for the firm's own transport emissions and the international freight carrying its imports are positive. The results therefore point to an increase in transport-related emissions alongside a decline in domestic emission intensity, while providing no precise evidence of a corresponding change in the main production-based emissions components.

Figure 8: The same components in levels



Notes. No denominator: every row is a level. **Everything the inventory counts is flat; the only emissions that rise are own transport and international freight.** Scale rises as well. Firm and year effects, full sample.

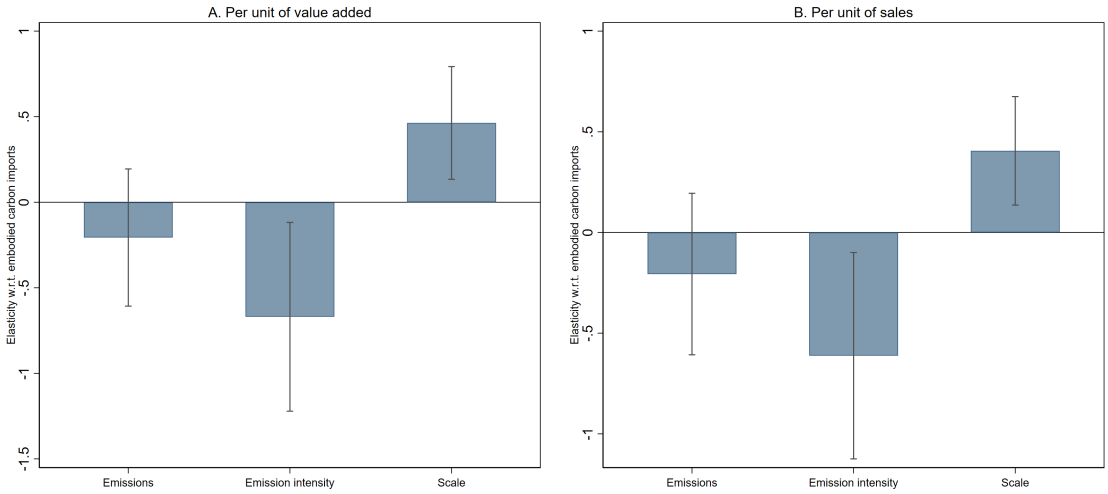
The distinction between emissions and emission intensity can be expressed through the accounting identity

$$\ln(E/VA) = \ln E - \ln VA,$$

where E denotes emissions and VA denotes value added. The elasticity of emission intensity is therefore the difference between the elasticities of emissions and value added. Estimating these components on a common sample and under the same specification makes it possible to assess how much of the intensity response reflects changes in emissions and how much reflects changes in value added. I use value added as the denominator, consistent with the firm-level decomposition in Cherniwchan et al. (2017). It measures emissions relative to the value generated

by the firm's domestic production, rather than relative to gross sales.

Figure 9: Does the firm get cleaner, or just bigger?



Notes. Each bar is the 2SLS elasticity of the stated outcome with respect to log embodied-carbon imports, on 14825 firm-years with firm, sector–year and municipality–year effects and standard errors clustered by firm, as in column (3) of Table 2, whose coefficient the emission-intensity bar reproduces. Because $\ln \mathrm{CO}_2 = \ln(\mathrm{CO}_2/VA) + \ln VA$ holds observation by observation and 2SLS is linear in the outcome, the emissions elasticity is the sum of the other two exactly; the adding-up is verified to 10^{-8} before the figure is drawn, which requires the sample on which all three are defined and so a smaller one than Figure 8, estimated at firm and year effects on the full sample. Panel A uses value added as the denominator and Panel B sales, so the emissions bar is common to both and only the split differs. Capped lines are 95 percent intervals, each from its own regression. The decomposition follows Barrows and Ollivier (2021).

Figure 9 decomposes the estimated change in emission intensity into its emissions and value-added components. Using the 14825 firm-years for which all three outcomes are defined, and the fixed effects in column (3) of Table 2, emission intensity falls by -0.670, value added rises by 0.463, and emissions change by -0.206. The emissions estimate is not statistically distinguishable from zero. The three estimates satisfy the accounting identity because they are calculated on the same sample and under the same specification. Just over two thirds of the decline in emission intensity is accounted for by the increase in value added, rather than a reduction in emissions. Using sales instead of value added as the denominator gives a similar decomposition, with an intensity change of -0.612 and a sales response of 0.406. The production-based intensity measure therefore improves, but the estimates provide no precise evidence that the firm's emissions themselves decline.

Value added and sales rise by nearly the same proportion, 0.463 and 0.406, respectively. Their similar responses imply little change in the aggregate share of intermediate inputs in sales, but do not rule out changes in the composition or origin of those inputs. One possible interpretation is that access to imported intermediates lowers marginal costs and allows firms to expand pro-

duction, as in Halpern et al. (2015); Amiti and Konings (2007). I do not identify this input-price channel directly. What the estimates establish is that domestic activity expands without a precisely estimated increase in on-site emissions. Emission intensity consequently falls, even though the results provide no precise evidence of a reduction in the firm's emissions in levels.

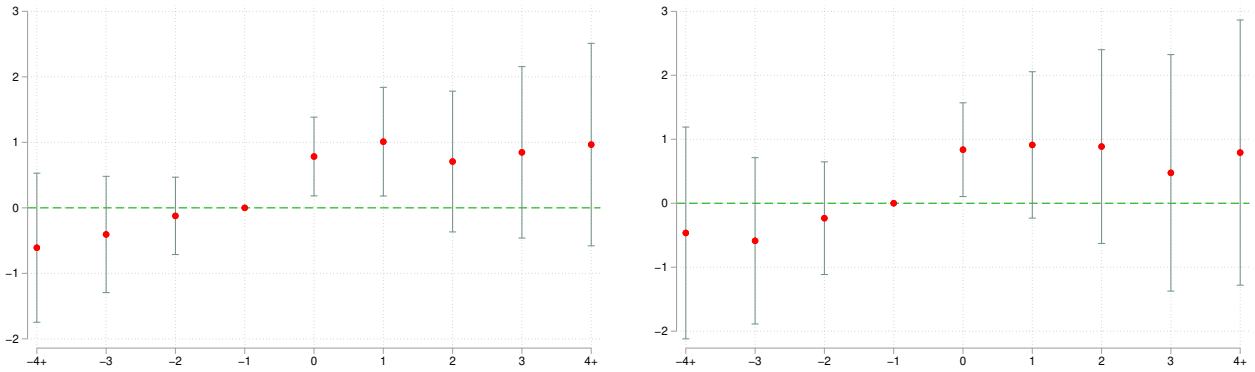
This finding complements Akerman et al. (2024), who use the same Swedish registers and attribute the relationship between imports and domestic emissions largely to cleaner technology. My results identify a different margin. The decline in production-based emission intensity is driven largely by the expansion of value added, while the emissions associated with the firm's own transport and the international freight carrying its imports increase. The two studies therefore speak to distinct components of the environmental consequences of importing: changes in domestic production and changes in the emissions associated with sourcing and moving foreign inputs.

7.2 Own transport emissions

The firm's own transport emissions provide one part of the transport-related response; international freight provides the other. Figure 10 traces the firm's own transport emissions over event time, alongside the production-based emission-intensity measure. From the year of the shock onwards, the two series move in opposite directions: production-based intensity declines, while own transport emissions increase, with the difference widening at longer horizons. This contrast is consistent with a shift in sourcing towards more distant suppliers, which can reduce emissions recorded at the production site while increasing the transport required to move intermediate inputs. The event-time pattern, however, does not by itself identify this specific sourcing response.

International freight is the larger transport-related margin in the estimates, but it is not recorded in the importing manufacturer's production account. It captures the emissions generated in moving imported intermediate goods from their country of origin to the importing firm. These emissions are treated separately from the firm's production emissions and are not assigned to the importing manufacturer in the same way as fuel burned at its own premises. Research on trade and the environment has paid comparatively less attention to emissions from moving goods than to emissions from producing them (Copeland et al., 2022). This distinction also matters for policy: the EU's carbon border adjustment mechanism covers the embedded emissions of specified imported goods, but does not generally include the international freight emissions associated with their transport. The freight component therefore remains outside the

Figure 10: Own transport emissions over event time



A. Transport CO_2 **B.** Transport CO_2 per unit of value added

Notes. Instrumented event-study coefficients β_m with $\beta_{-1} = 0$, 95 percent confidence intervals clustered by firm, balanced 2006–2014. Panel A is transport CO_2 in levels; Panel B scales it by value added, so the reversal is not an artefact of output growing. In both, the sign flips at the shock and the on-impact and year-one effects are significant; from year two the estimates stay positive but the intervals are wide, so the figure establishes a sign reversal, not a precisely estimated long-run path.

mechanism's carbon-cost calculation. I construct international freight emissions as

$$E_{it}^{\mathrm{FR}} = \sum_{c} M_{ict}\, d_c\, \varphi_{m(c)}, \tag{7}$$

The measure combines the net mass of firm i's imports from country c, M_{ict}, the distance travelled, d_c, and the emission factor associated with the assigned transport mode, $\varphi_{m(c)}$. The emission factors are expressed in grams of CO_2 per tonne-kilometre. The distance measure and the assignment of transport modes are constructed as described in Appendix Q. The approach to measuring freight emissions follows Cristea et al. (2013), while the modal emission factors are taken from the GLEC Framework v3 and the IMO Fourth Greenhouse Gas Study. Because import weights are measured as net mass, the calculation excludes packaging and other weight carried alongside the imported goods. The resulting measure therefore does not capture the full weight transported.

7.3 The net footprint

The total footprint combines emissions from domestic production, international freight and the foreign production of imported inputs:

$$E_{it}^{\mathrm{ACC}} = E_{it}^{\mathrm{DOM}} + E_{it}^{\mathrm{FR}} + E_{it}^{\mathrm{FGN}}.$$

But this total is not the additional carbon attributable to offshoring. The imported inputs would have generated production emissions even if they had been produced in Sweden. The relevant comparison is therefore with the emissions that domestic production of the same import basket would have generated. Section 7.4 constructs this counterfactual, denoted by C_{it}^{swe}, by valuing the firm's import basket at Swedish emission intensities. The additional emissions associated with offshoring are consequently measured relative to this domestic-production benchmark:

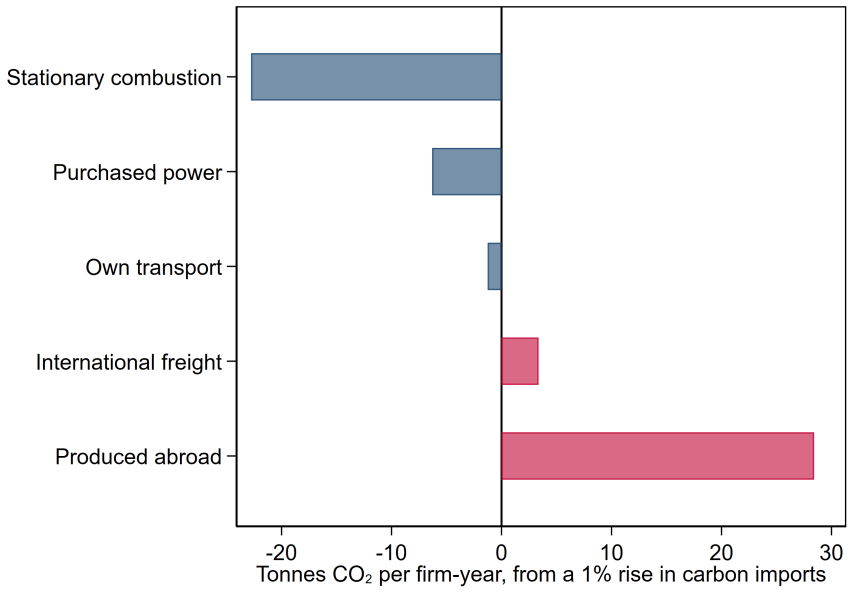
$$E_{it}^{\text{NET}} = E_{it}^{\text{DOM}} + E_{it}^{\text{FR}} + \underbrace{\left(C_{it}^{for} - C_{it}^{swe}\right)}_{\text{origin-dirtiness margin}}, \tag{8}$$

The third term is the origin-dirtiness margin estimated in Section 7.4. Its magnitude matters for the accounting. At the sample median, the carbon embodied in the imported basket is 1.64 times the emissions that the same basket would have generated at Swedish emission intensities. The excess attributable to the origin-dirtiness margin is therefore only the difference between these two quantities. Counting all foreign embodied emissions as additional emissions from offshoring would overstate this margin by a factor of approximately 2.56. More generally, a decline in E^{DOM} need not imply a decline in the net footprint if international freight and the origin-dirtiness margin increase.

Figure 11 expresses the estimated responses in tonnes of CO_2, allowing the different emissions components to be compared in a common unit. Each bar converts the estimated elasticity into a change in emissions evaluated at the sample mean. The first four bars report changes in emissions recorded in the firm's accounts. The fifth reports the gross carbon embodied in imported inputs, valued at the emission intensity of their countries of origin. It is therefore an accounting measure of foreign production emissions, not the additional emissions attributable to offshoring.

That fifth bar is an accounting entry. The elasticity of foreign emissions with respect to carbon imports is one by construction, since the regressor and that outcome are the same variable, and I report it only because leaving it out would make the figure add up to a saving the accounting does not support. It plots C^{for}, the gross quantity, so the figure overstates the addition by the Swedish counterfactual it does not net out. Equation (8) is the quantity the argument rests on, and the median ratio of 1.64 is the size of the gap between them. What the figure shows is the domestic reduction set against the carbon that reappears abroad and in transit. The scope of E^{NET} also needs to be clear. It does not include emissions embodied in inputs that the firm purchases domestically, either in its observed production or in the counterfactual. Nor does it include emissions downstream of the firm's sales or embodied in its capital goods. It is therefore not a comprehensive consumption-based footprint of the firm. Rather, it measures the domes-

Figure 11: The components in tonnes



Notes. Each bar is the change in tonnes implied by the estimated elasticity for a firm at the sample mean, so the components add instead of being compared as percentages. The fifth bar, carbon embodied in imports, is the regressor itself expressed in tonnes: carbon embodied in imports (in kg) and the regressor are both built as import value times the origin's embodied intensity, so that bar and the total are close to arithmetic, not estimates. Read the first four bars, which are emissions the firm reports.

tic production, freight and origin-dirtiness emissions associated with the firm's imported input basket, relative to producing that basket in Sweden.

7.4 The margins of embodied carbon

A firm that shifts a given import basket towards more carbon-intensive origins increases the emissions embodied in its imports, relative to what the same basket would have generated in Sweden. This is the origin-dirtiness margin at the heart of the pollution-haven hypothesis. Yet the empirical literature has found limited evidence that firms systematically source from less regulated, more polluting locations. This section revisits that result by separating origin choice from the volume and composition of imports, and by examining whether the pooled response conceals differences across firms.

A natural test of the pollution-haven hypothesis is to split the instrument according to the environmental policy of the exporting origin. In this setting, however, the origin-specific instruments do not provide a usable first stage. As Appendix G explains, firms' exposure to the different origin groups is constructed from the same baseline product shares. Splitting the foreign export-supply shock by origin group therefore fragments the variation that identifies the pooled

instrument, leaving insufficient predictive power in the separate instruments. I consequently use the pooled instrument to examine a different question: which component of embodied-carbon imports responds to the shock?

I instead use the pooled instrument to distinguish three margins through which embodied carbon in imports can change: the volume of imports, the composition of the imported basket, and the emission intensity of the origins supplying it. These margins can be separated using the following accounting identity:

$$\ln C_{it}^{for} = \underbrace{\ln M_{it}}_{\text{volume}} + \underbrace{\ln\left(C_{it}^{swe}/M_{it}\right)}_{\text{product mix}} + \underbrace{\ln\left(C_{it}^{for}/C_{it}^{swe}\right)}_{\text{origin dirtiness}},$$

where C^{swe} values the firm's actual import basket at Swedish emission intensities. The product-mix term captures changes in the composition of imports, evaluated at Swedish emission intensities. The origin-dirtiness term captures the difference between the carbon intensity of the actual sourcing origins and the Swedish benchmark for that same basket. It is this third term that provides the test of sourcing from relatively more carbon-intensive origins.

I regress each margin directly on the instrument rather than instrumenting it separately. The object of interest is the reduced-form response of each component to the foreign export-supply shock. Since the three margins sum to $\ln C_{it}^{for}$ by the accounting identity above, their reduced-form coefficients also add up to the reduced-form coefficient for embodied-carbon imports, provided they are estimated on the same sample and with the same specification. This approach does not require a separate first stage for each component. Its causal interpretation, however, rests on the same exogeneity condition as the main design.

I also examine two additional margins. The first is domestic technique, which the shock should not affect and which therefore serves as a placebo. The second is international freight, whose emissions arise in transit rather than at the production site. Prediction 2 implies that the ratio of freight emissions to the carbon embodied in the goods transported should not respond to the shock unless the shock shifts sourcing towards more distant origins. Freight emissions may nevertheless rise in levels, as shown in Section 7. The decomposition therefore allows me to distinguish an increase in the volume of carbon imports from changes in product composition, origin dirtiness and the transport required to move those imports.

Panel A of Figure 12 reports the reduced-form responses of the three margins pooled across firms. The volume margin accounts for approximately four fifths of the total response in embodied-carbon imports, consistent with Prediction 1. The product-mix response is substantially smaller, as the prediction implies. The origin-dirtiness estimate, by contrast, is not

statistically distinguishable from zero. This reproduces the pooled null reported in much of the previous literature. However, the point estimate for origin dirtiness is similar in magnitude to that for product mix. The absence of statistical significance should therefore not be read as evidence that the estimated origin-dirtiness response is economically smaller; the distinction between the two estimates lies primarily in their precision.

The scope for reallocating imports across origins may depend on the breadth of a firm's product portfolio. Firms purchasing more distinct products have a wider set of product categories across which sourcing patterns can potentially adjust, although they may also switch suppliers within a given product. The pooled estimates do not reveal whether the origin-dirtiness response differs with this initial product scope. I therefore split firms according to the number of distinct products they purchased in the first sample year, using this pre-determined measure to define the groups in Panel B of Figure 12.

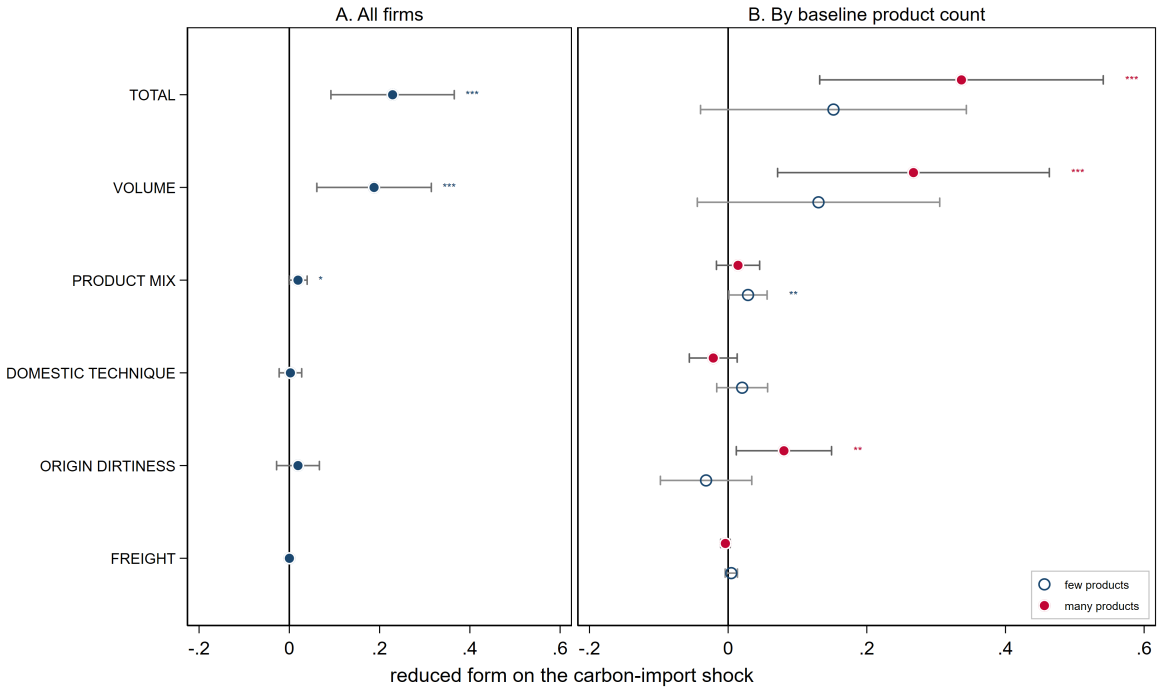
The split follows Prediction 3, which concerns how the two composition margins respond across firms with different initial product scope. Origin dirtiness is statistically distinguishable from zero among firms that initially bought many products, but not among firms that bought few. Product mix displays the opposite pattern: its estimated response is larger among narrow-scope firms. This pattern is inconsistent with a simple account in which broad-scope firms respond more strongly to the shock across all margins. Instead, it is consistent with the model's distinction between reallocating the origins from which existing products are sourced and changing the composition of the product basket.

7.5 Trade and ownership

A common approach in the pollution-haven literature is to examine whether multinational firms shift production towards countries with less stringent environmental regulation. This focuses attention on foreign direct investment and the location of foreign affiliates. Firm-level trade records allow a different distinction: whether the carbon embodied in imports responds through purchases from countries where the importing firm owns a plant, or through purchases from other foreign suppliers. I use this distinction to assess how much of the sourcing response operates within multinational production networks rather than through arm's-length trade.

Table 3 separates embodied-carbon imports into purchases from countries where the importing firm owns a plant and purchases from all other foreign suppliers. For arm's-length imports, the IV estimate is -0.676. The first-stage statistic is 8.61, so I report the Anderson–Rubin confidence set, which ranges from -1.93 to -0.32 and excludes zero. This provides evidence of a negative response under inference that remains valid with weak identification, conditional on

Figure 12: The five margins of embodied carbon



Notes. Panel A pools all firms. Panel B splits them on the number of distinct products the firm bought in its first sample year, fixed before any shock arrives. Each bar is a reduced-form coefficient of the margin on the carbon-import shock, for the five margins of the identity in Section 7.4; they sum to the total by construction. Fixed effects as in column (3) of Table 2. Standard errors clustered by firm. Stars denote significance at 10, 5 and 1 percent.

the instrument's exclusion restriction. The own-affiliate estimate is much less informative: its first-stage statistic is 0.16, and its Anderson–Rubin confidence set covers the entire searched grid. This does not establish that the own-affiliate response is zero. Nor does the comparison establish that the arm's-length coefficient is statistically different from the own-affiliate coefficient: the two estimates come from separate just-identified regressions, and their difference is not itself tested here.

The regression results identify a response in arm's-length sourcing but leave the own-affiliate response imprecisely estimated. I therefore consider three additional pieces of evidence on the quantitative importance of ownership in firms' sourcing patterns. They address different questions: how much imported carbon enters through countries where the firm owns a plant, whether sourcing from such countries is associated with higher embodied carbon, and how much of the overall sourcing response can plausibly operate through multinational production networks.

The first piece of evidence is descriptive. Figure 13 classifies manufacturing imports according to whether the importing firm owns a plant in the country of origin. Only 2.0 percent of embodied-carbon imports arrive from countries in which the importing firm has an affiliate. This

Table 3: Multinational status and the sourcing margins

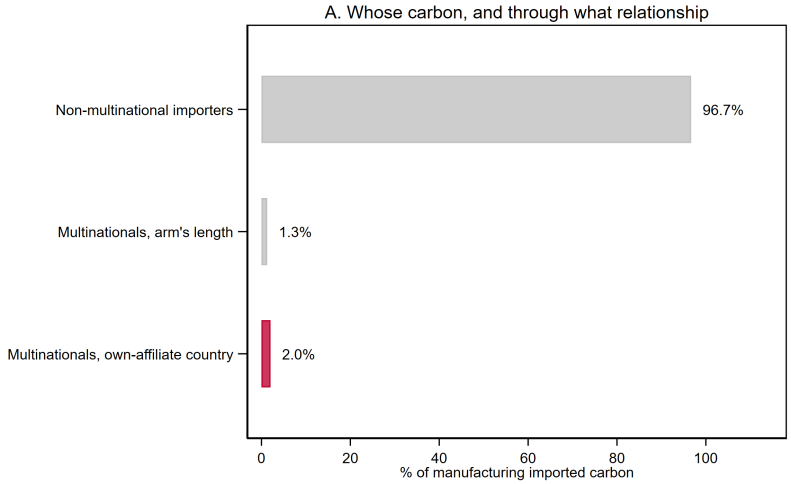
	N	OLS	2SLS	KP F	AR 95% set
Panel A. Groups and sourcing margins					
All firms	15,175	-0.025	-0.524***	13.92	[-1.12, -0.23]
			(0.183)		
MNE parents	322	-0.048	-0.249	6.86	[-1.85, 0.89]
			(0.464)		
Non-MNE	14,808	-0.025	-0.528***	13.20	[-1.16, -0.23]
			(0.188)		
Own-affiliate imports	11,587	0.001	4.379	0.16	$\supseteq [-10, 10]$
			(11.121)		
Arm's-length imports	11,587	-0.014	-0.676**	8.61	[-1.93, -0.32]
			(0.264)		
Panel B. Formal interaction, fixed effects crossed with MNE status					
Non-MNE			-0.528***	6.59	
			(0.188)		
MNE parents			-0.249	6.59	
			(0.456)		
Difference			0.279	6.59	
			(0.493)		

Notes. Firm and year effects. The Kleibergen–Paap rk Wald F is reported beside each row; where it falls below 10 the conventional interval is not valid and the Anderson–Rubin set, whose coverage does not depend on the first stage, is reported instead. Own-affiliate and arm's-length sourcing are the *same* firms and the *same* instrument with a different regressor: two just-identified regressions, **not** a decomposition. The own-affiliate row has $F = 0.16$ and an AR set covering the whole search grid, and is reported to show that this margin is uninformative, not null. The Anderson–Rubin sets are obtained by inverting the test on 81 points spanning $[-10, 10]$, so a set written $\supseteq [-10, 10]$ is one the test rejects nowhere on that grid; no value in it is excluded and no value outside it was examined. Panel B crosses the fixed effects with multinational status, so the difference is tested. It is 0.279 with a standard error of 0.493.

share does not rely on the instrument or on an identifying assumption. It measures the maximum share that can be classified as potentially intra-firm using the ownership-country match. It is an upper bound because the trade register identifies the country of origin but not the transaction counterparty: purchases from unrelated suppliers in countries where the firm owns a plant are also included. The actual share of intra-firm imports must therefore be smaller.

The second piece of evidence asks whether imports from countries where a firm owns a plant are more carbon-intensive than its imports from other origins. Table 4 makes this comparison within firm–years, across the origins from which each firm imports, while origin–year fixed effects absorb factors common to each country in a given year. Imports from countries where the firm owns a plant are associated with higher embodied carbon (0.561), but also with a similar increase in import value (0.580). Since $\ln(C/M) = \ln C - \ln M$, the implied difference in the carbon intensity of imported purchases is small, at -0.019. The estimates therefore provide little evidence that imports from affiliate countries are more carbon-intensive than imports from other origins. This is a conditional within-firm-year association, not a causal effect of ownership: firms

Figure 13: Whose carbon, and through what relationship



Notes. All manufacturing imported carbon, 2008–2014. Only 2.0 percent enters through a country where the importing firm owns a plant, and that is an *upper bound*: the register records the origin country, not the counterparty, so arm's-length purchases from third parties in that country are counted here.

Table 4: Does owning a plant abroad change what a firm sources from there?

Specification	N	Coefficient
Carbon sourced (firm-year FE only)	159,839	1.076***
		(0.204)
Carbon sourced (+ origin-year FE)	159,610	0.561***
		(0.184)
Import value (+ origin-year FE)	159,610	0.580***
		(0.181)
Carbon per krona (+ origin-year FE)	159,610	-0.019
		(0.023)

Notes. Firm × year × origin. Within a firm-year, across the origins it buys from, the regressor is an indicator for the firm owning a plant in that origin. Origin-year effects absorb anything common to a country in a year. **Ownership raises how much a firm sources from an origin, not how dirty it is.** Carbon sourced rises 0.561 and import value 0.580; since $\ln(C/M) = \ln C - \ln M$, carbon per krona is their difference, -0.019 with a standard error of 0.023. That interval rules out even a six percent change in sourcing dirtiness.

choose where to establish affiliates, and the comparison does not make that choice exogenous.

The third piece of evidence examines whether opening a foreign affiliate is followed by a change in the Swedish parent's emission intensity. Figure I.1 traces the response around a firm's first affiliate in a new country. Because treatment is staggered and absorbing, conventional two-way fixed-effects event-study estimates can be difficult to interpret when treatment effects vary across cohorts or over time. The setting also includes firms that never open an affiliate, allowing me to use the Callaway–Sant'Anna estimator with never-treated firms as the comparison group. I report estimates from this approach alongside those from Sun–Abraham and conventional two-way fixed effects. The pre-treatment coefficients are jointly indistinguishable from zero ($F = 1.06$,

$p = 0.348$), and none of the estimated post-entry effects is statistically distinguishable from zero. Appendix I extends the analysis across the event-time horizons supported by the panel and discusses the differences in the firms contributing to estimates at shorter and longer horizons.

The interpretation of these null estimates depends on the horizon over which the response can be observed. Relocating production may take time, and the event-study estimates are based on roughly 1,200 treated firm–years in a panel ending in 2014. They provide no statistically detectable evidence of a change in the parent's emission intensity over the observed post-entry horizons. They cannot, however, rule out responses that emerge over a substantially longer period, such as fifteen years after entry. The result is therefore limited to the period covered by the data: it does not support the view that foreign-affiliate entry accounts for the observed changes in firms' emission intensity during this decade, but it does not exclude longer-run effects of multinational expansion.

A split of the main estimate by multinational status is not informative enough to establish whether the response differs between multinational and non-multinational firms. The point estimate for multinational parents is -0.249, compared with -0.528 for other firms. However, the multinational first-stage statistic is 6.86, and the Anderson–Rubin confidence set ranges from -1.85 to 0.89, including zero. The subgroup contains only 322 multinational firm–years. As with the split by origin group discussed in Section G, dividing the shift-share instrument by firm type leaves too little identifying variation for a precise subgroup estimate. The comparison therefore establishes neither a multinational-specific response nor a difference between the two groups. This is why I rely instead on the ownership-country share, the within-firm-year comparison across origins, and the affiliate-entry event study to examine the role of multinational production networks.

Taking stock, the evidence points to a distinction between arm's-length trade and multinational ownership. The arm's-length sourcing response is statistically distinguishable from zero under weak-instrument-robust inference, whereas the response in own-affiliate imports is too imprecisely estimated to draw a comparable conclusion. At most 2.0 percent of embodied-carbon imports arrive from countries where the importing firm owns a plant. The within-firm-year comparison provides little evidence that imports from these origins are more carbon-intensive, while the affiliate-entry event study detects no statistically distinguishable change in the parent's emission intensity over the observed horizons. The evidence therefore suggests that ownership-based measures capture only a limited part of the sourcing patterns documented here. Policies focused on the carbon embodied in imported goods may address a broader margin than those targeted solely at multinational ownership, although this paper does not estimate the effects of

such policies directly.

8 Conclusion

This paper examines how offshoring changes the emissions recorded at home when imports are measured by their embodied carbon rather than their monetary value. Using data on Swedish manufacturing firms and a shift-share instrument based on foreign export-supply shocks, I estimate that a 10 percent increase in embodied-carbon imports reduces domestic CO_2 emissions per unit of value added by 6.7 percent. Under the identifying assumptions of the design, this represents a decline in production-based emission intensity, not necessarily a reduction in absolute emissions or in the emissions embodied in the firm's production and imported inputs. Consistent with this distinction, the estimated effect on emissions in levels is statistically indistinguishable from zero.

The decline does not reflect a cleaner energy mix. It is associated with lower energy use per unit of value added, rather than a shift towards less carbon-intensive fuels: the estimated fuel shares and emissions per unit of energy do not change detectably. This is consistent with firms already relying predominantly on electricity, leaving limited scope for further fuel switching. Value added also rises, so part of the decline in emission intensity comes from growth in the denominator rather than a reduction in emissions in the numerator.

The results qualify the conventional pollution-haven account of offshoring. The volume margin accounts for approximately four fifths of the reduced-form response in embodied-carbon imports, while the average response of sourcing-origin carbon intensity is not statistically distinguishable from zero. This pooled result masks a pattern across firms: the origin-dirtiness estimate is statistically distinguishable from zero among firms with broader baseline product scope, but not among those with narrower scope. This pattern is consistent with the model's fixed-cost mechanism. Reaching a new origin entails a cost, so firms already sourcing across a wider range of products and origins have more scope to reallocate purchases across suppliers. The evidence is therefore consistent with origin substitution among firms with broader initial sourcing scope, rather than a uniform shift towards dirtier production locations.

Finally, the environmental consequences of foreign sourcing extend beyond the emissions recorded in domestic production. The carbon embodied in imported goods is greater than the emissions that producing the same basket at Swedish emission intensities would generate. The estimates also indicate increases in firms' own transport emissions and in the international freight associated with their imports. These transport emissions are not fully captured in the production-

based accounts of Swedish manufacturing. The distinction matters: the carbon embodied in an imported input reflects emissions from producing that good abroad, whereas freight emissions arise from moving it to the importing firm. The ownership evidence further suggests that imports from countries where the importing firm owns a plant account for at most 2 percent of embodied-carbon imports. This is an upper bound based on matching origin countries to affiliate ownership, not a direct measure of intra-firm trade. It nevertheless indicates that an approach focused exclusively on foreign affiliates would leave most of the imported-carbon flow outside its scope.

The results reveal a gap between the emissions recorded in domestic production accounts and the wider environmental consequences of foreign sourcing. Swedish manufacturing firms reduce their production-based emission intensity, but the estimates do not establish a corresponding reduction in their absolute emissions. At the same time, the emissions associated with firms' own transport and the international freight carrying their imports increase. For policy, this distinction suggests that approaches focused exclusively on the ownership of foreign affiliates may overlook much of the carbon embodied in imported inputs. Policies that account for the carbon content of imports address a broader margin, although this paper does not estimate their effects. Quantifying the aggregate gains from such policies requires a general-equilibrium analysis of trade and emissions; the accounting exercise here is partial equilibrium throughout (Farrokhi and Lashkaripour, 2021; Nordhaus, 2015). The contribution of this paper is to document, at the firm level, how changes in sourcing affect domestic production-based emissions and the emissions associated with producing and transporting imported inputs.

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Appendix

A Variable definitions and summary statistics

Table A.1 defines every variable that enters the estimates, with its source register and, where the variable is constructed, its formula.

Table A.1: Variable definitions

Variable	Definition and source
Outcome	
CO_2, tonnes	Firm's own stationary combustion, from fuel quantities by type valued at fuel-specific emission factors. SCB energy register.
Value added, SEK	Output less intermediate inputs. SCB firm register.
Emission intensity	$\ln(CO_2/\text{value added})$. The main results also report sales and employment as alternative denominators.
Endogenous regressor and instrument	
Embodied carbon imports, C^{for}	$\sum_{c,k}$ trade value$_{ickt} \times e_{ckt}$, where e is the EXIOBASE total (direct plus upstream) CO_2 intensity of product k in country c, obtained from the Leontief inverse $S(I-A)^{-1}$.
Shift-share instrument	$\sum_k s_{ik,2005}\,\Delta\text{WES}_{kt}$: pre-sample product shares times the change in world export supply of k, leaving out Sweden.
Decomposition margins	
Import value, M	Customs value of the firm's imports. Volume margin.
Swedish-valued basket, C^{swe}	The same import basket valued at Swedish emission intensities. Product mix is $\ln(C^{swe}/M)$.
Origin dirtiness	$\ln(C^{for}/C^{swe})$: how much dirtier the chosen origins are than Sweden would be for the same basket.
Domestic technique	Emissions per unit of output on the firm's own production. Placebo margin.
Freight, E^{FR}	$\sum_c M_{ict}\, d_c\, \varphi_{m(c)}$: import weight in tonnes times CEPII distance times a modal CO_2 factor. Appendix Q.
Moderators and controls	
Sector carbon intensity	Emissions per unit of value added at the 3-digit sector level, split at the manufacturing median.
EU ETS	Firm holds emission allowances in the Union registry in a given year.
Carbon tax rate	Firm's effective rate: statutory rates by fuel weighted by the firm's own fuel mix. Endogenous to the sourcing decisions under study and reported only as a specification check.
Multinational parent	Firm reports at least one foreign affiliate in the FDI register.
Own-affiliate imports	Imports from an origin in which the firm reports an affiliate. The register records origin country, not counterparty, so this is an upper bound on intra-firm trade.
Productivity	Log TFP, estimated following Wooldridge (2009).

Notes. Swedish manufacturing firms importing intermediate inputs, 2006–2014, with 2005 held out to form the instrument's baseline shares. All monetary variables are deflated to 2010 SEK. The EXIOBASE intensity e is published per million euro of final demand, so C^{for} is a mixed unit; Appendix Q gives the conversion to kilograms used wherever a physical quantity is required.

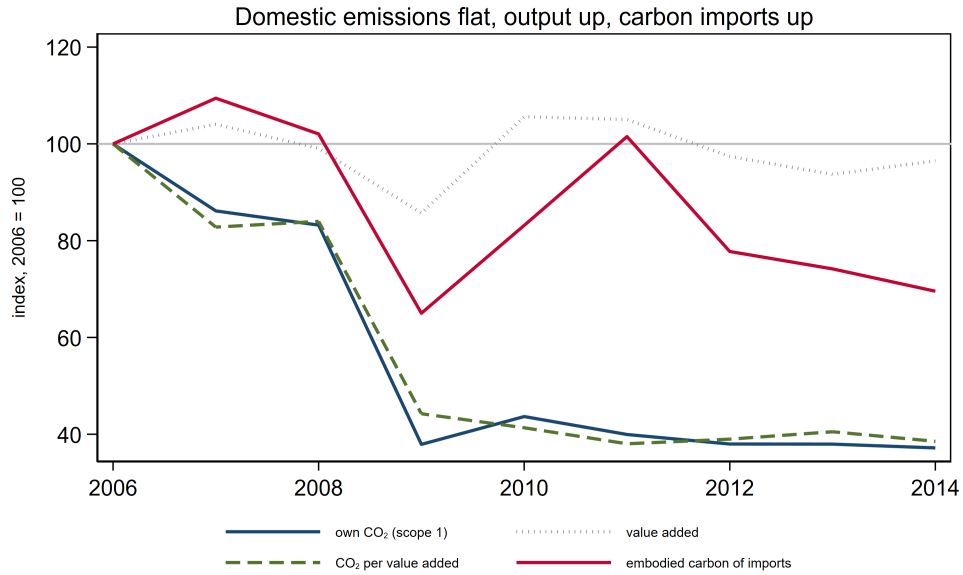
Table A.2: Summary statistics, estimating sample

Variable	N	Mean	SD	p10	Median	p90
Firm characteristics						
Employees	15,749	168.82	767.05	13.00	47.00	303.00
Sales, MSEK	15,749	643.71	4332.29	21.30	95.66	879.00
Value added, MSEK	15,749	171.14	1145.71	7.60	28.81	234.66
Energy supply, GJ	15,749	280845.90	2346319.20	1151.28	7562.60	138920.22
Distinct EXIOBASE products, first	15,749	5.06	4.21	1.00	4.00	11.00
Distinct origin countries, first y	15,749	6.99	7.18	1.00	5.00	16.00
Emissions						
CO2 scope 1, tonnes	15,749	5224.60	93386.99	6.90	76.48	1960.04
CO2 scope 2, tonnes	15,749	1152.79	6889.71	8.45	71.81	1022.79
CO2 per SEK value added, kg	15,749	0.02	0.11	0.00	0.00	0.03
CO2 per SEK sales, kg	15,749	0.00	0.04	0.00	0.00	0.01
CO2 per employee, kg	15,696	12522.98	85806.47	160.57	1851.61	16581.51
Trade and carbon imports						
Intermediate imports, MEUR	15,749	10.06	77.80	0.00	0.62	13.53
Embodied carbon of imports, tonnes	15,749	7066.54	45992.71	1.41	400.66	10471.59
Import intensity, kg CO2 per EUR	15,749	0.81	0.85	0.30	0.61	1.46
Swedish counterfactual, kg CO2 per	15,749	0.43	0.30	0.20	0.37	0.77
Foreign / Swedish intensity	15,749	2.18	3.02	0.87	1.64	3.67
Freight CO2, tonnes	15,749	400.09	2460.29	0.01	9.25	572.54
Freight as % of imported carbon	15,749	4.97	7.83	0.21	2.39	12.04
Basket churn vs own first year	15,749	0.26	0.37	0.00	0.04	1.00

Notes. Estimating sample, 2005–2014. Levels, not logs, so that magnitudes can be checked against external benchmarks: EXIOBASE embodied intensities for manufacturing run roughly 0.3–1.5 kg CO_2 per euro. Product and origin counts are taken in the firm's first sample year, before any adjustment to the shocks used for identification. Basket churn is the share of the firm's import value in products it did not buy in its own first sample year.

Figure A.1 plots emissions and output over the estimating sample year by year, which places the firm-level results against the aggregate they sit inside. Emissions fall while output rises across the window, which is the aggregate pattern the paper sets out to account for. Table A.3 then compares the firms that enter the estimating sample with those that do not, since the sample is restricted to importers of intermediate inputs and that restriction is not random: entering firms are larger and more carbon-intensive, which bounds the population the estimates describe.

Figure A.1: Emissions and output in the estimating sample



Notes. Totals over the estimating sample, by year.

Table A.3: Estimating sample against manufacturing firms outside it

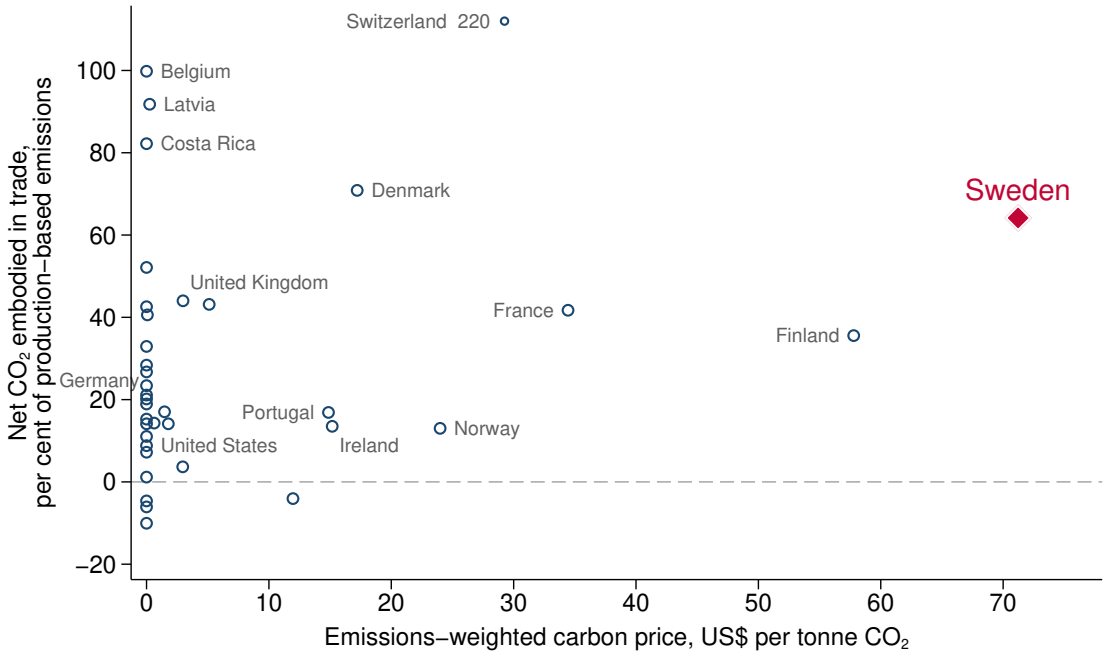
Variable	In sample	Not in sample	Norm. diff.
Employees	168.82	45.29	0.213
Sales, MSEK	643.71	119.62	0.162
Value added, MSEK	171.14	30.45	0.164
CO2 scope 1, tonnes	5224.60	1173.08	0.055
CO2 per SEK value added, kg	0.02	0.01	0.021
Energy supply, GJ	280845.90	61839.86	0.119
Ever in the EU ETS	0.06	0.01	0.257

Notes. Normalised differences (Imbens and Rubin, 2015), not t-tests: with roughly 40,000 firm-years every difference is significant at 1%, so significance stars carry no information about whether the two groups resemble one another. A normalised difference above 0.25 in absolute value conventionally indicates substantial imbalance. The estimating sample is larger and more energy-intensive by construction, since it consists of importing manufacturers. Emission *intensity* — the outcome — is close to balanced. EU ETS participation is the one variable that crosses the threshold; Section 6.3 reports the main specification excluding ETS firms.

B Carbon pricing and embodied carbon across the OECD

The introduction motivates the Swedish setting with two numbers: an emissions-weighted carbon price of about $71 per tonne in 2020, and net carbon imports equal to 64 percent of production-based emissions. Figure B.1 places those two figures against the other OECD economies, so that how unusual the combination is can be judged rather than asserted.

Figure B.1: Sweden prices carbon hardest and still imports a great deal of it



Notes. OECD economies in 2020. The horizontal axis is the emissions-weighted carbon price, which multiplies the statutory rate by the share of a country's CO_2 it covers; the vertical axis is net CO_2 embodied in trade as a percentage of production-based emissions, so a positive value means a country consumes more carbon than it produces. Switzerland is plotted at the axis maximum and labelled with its true value. National accounts, constructed separately from the firm data used below. *Sources:* Global Carbon Project and the World Carbon Pricing Database, retrieved from Our World in Data; see ourworldindata.org/grapher/share-co2-embedded-in-trade and ourworldindata.org/grapher/emissions-weighted-carbon-price. Both series are documented at ourworldindata.org/consumption-based-co2

C Environmental policy stringency

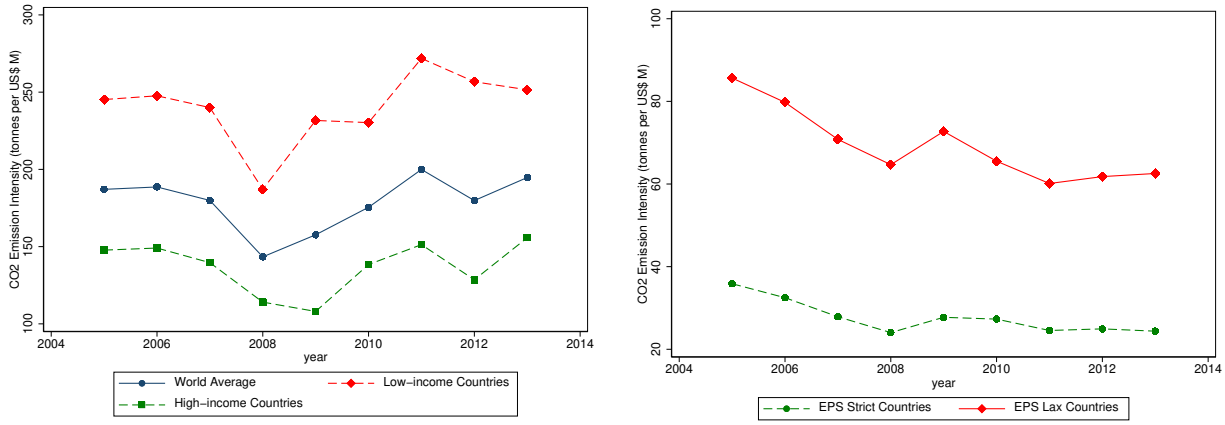
Origins are grouped by the OECD Environmental Policy Stringency index, a composite of roughly fifteen market and non-market instruments covering forty economies. I average it over 2005–2014 and measure each origin as a distance from Sweden's own value. About 80 percent of countries fall below Sweden, so a binary lax indicator separates almost nothing; I therefore cut the distance into four groups. Within half a mean deviation of Sweden is strict, half to one and a half is quite lax, and beyond that is very or extremely lax, giving a rank in $\{0, 1, 2, 3\}$.

Figure C.2 sets each origin's whole-economy carbon intensity against that grouping. Mean intensity rises across all four groups, from 109 tonnes of CO_2 per million euro of output among the strict countries to 463 among the extremely lax. Across individual countries the spread is wider still: Switzerland is the cleanest and Russia the dirtiest, a ratio of 14.5 to one, and Copeland et al. (2022) report a similar order of magnitude on different data.

The grouping is a crude summary and the figure shows where it fails. Brazil and Turkey sit in the extremely lax group and are comparatively clean, while Greece is only quite lax and

is the sixth dirtiest country in the sample. Energy mix, industrial composition and income all move with policy stringency, and none of them is separated here. That is why the classification describes the origins and identifies nothing: splitting the instrument by group leaves no first stage, as Table G.1 reports, and the policy question is answered instead through the origin-dirtiness margin in Section 7.4. What this appendix establishes is weaker and sufficient: origins differ enough in carbon intensity that where a firm buys can matter for the carbon embodied in what it buys.

Figure C.1: Trend of emission intensity by income groups and level of environmental laxity

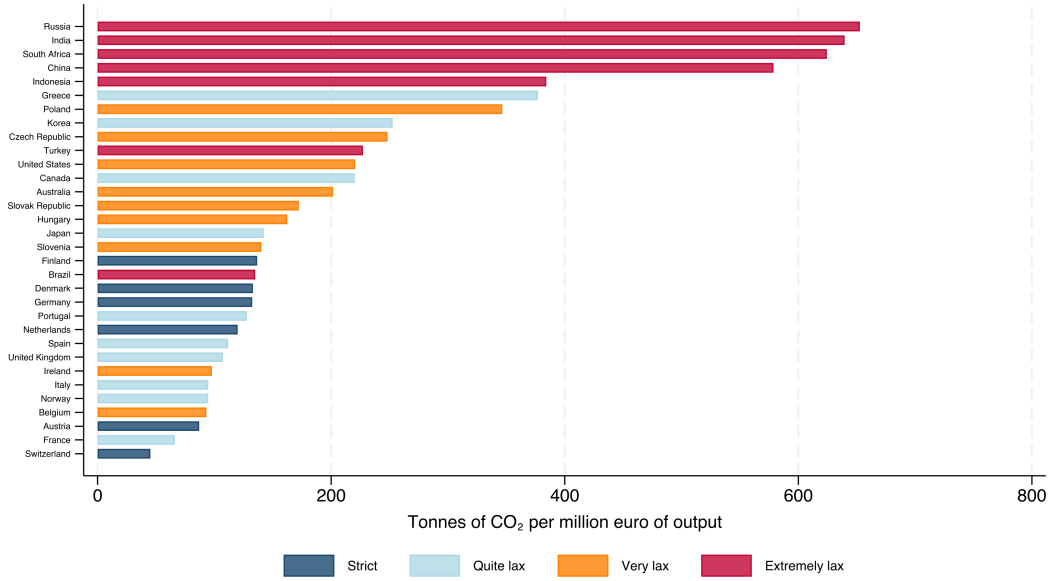


A. Low vs High Income Countries

B. EPS Lax vs EPS Strict Countries

Note: This figure plots the yearly CO_2 emission intensity (in tons per millions of US$). Low income countries includes lower middle income countries identified by the World Bank. EPS Lax and Strict countries are identified using the EPS index (see Appendix C).

Figure C.2: Emission intensity and environmental policy stringency



Notes. Each bar is a country's whole-economy CO_2 emissions divided by its output, in tonnes per million euro, averaged over 2006–2014 and summed across all products. Colours give the four-way policy grouping described in the text, built from the OECD Environmental Policy Stringency index averaged over 2005–2014 and measured as a distance from Sweden. The grouping is a country characteristic rather than a firm-level treatment, and nothing in the paper's identification rests on it. *Sources:* EXIOBASE 3 and the OECD.

D The aggregate decomposition

Decomposition of Swedish Emission. To understand the role of trade on emissions, I decompose the manufacturing CO_2 emission into three potential channels: (i) changes in the size of the manufacturing sector – scale effect, (ii) changes in emissions due to technology used in production within industries – technique effect, and (iii) changes in the mix of sectors towards less pollution-intensive industries – composition effect. [7] More formally, the CO_2 emissions from the manufacturing sector in year t, E_t can be written as follows:

$$E_t = \sum_j E_{i,t} = \sum_j v_{i,t} z_{j,t} = V_t \sum_i \theta_{j,t} z_{j,t} \tag{9}$$

where $j = 1, \ldots, n$ indexes manufacturing industries (defined at SIC 4-digits). V_t is the manufacturing monetary value of output, $\theta_{j,t}$ is each industry's share in total output $\left(\theta_i = v_{j,t}/V_t\right)$, and $z_{j,t}$ is each industry's emissions coefficient measured as the amount of pollution per monetary

[7] My working assumption, as motivated by Copeland et al. (2022) and Choi et al. (2023) is that an increase in physical unit of output (and monetary value of output) does not scale 1-to-1 with emissions because, firm adjust their production processes which can affect emissions and value added. Subsequently, I follow the decomposition exercise of Choi et al. (2023); Copeland et al. (2022); Ustyuzhanina (2022).

value of output in industry j $\left(z_{j,t} = p_{j,t}/v_{j,t}\right)$. Thus, the total CO_2 emissions in a particular year is the scale of the sector times weighted emission per monetary value of output ($V_t \sum_i \theta_{j,t} z_{j,t}$) – where θ and z are $j \times 1$ vectors containing the output shares of each of the j sectors and their emissions intensities, respectively. Equation 9 can be rewritten in a vector notation form as $E = V\boldsymbol{\theta}'z$. By applying a total differentiation on the vector notation, I obtain:

$$dE = \underbrace{\boldsymbol{\theta}' z dV}_{scale} + \underbrace{V z' d\boldsymbol{\theta}}_{composition} + \underbrace{V\boldsymbol{\theta}' dz}_{technique} \tag{10}$$

The three terms on the right-hand side in equation 10 can be interpreted as the three primary channels or effects that explain the change in total pollution over time, namely, the scale effect, the composition effect, and the technique effect. Figure 2 reports the decomposition, indexing aggregate manufacturing CO_2 to 1 in 2006, the first year the energy data exist. The construction follows Choi et al. (2023), who combine Levinson (2009) with Melitz and Polanec (2015). Line (1) varies only the scale of the sector, holding the industry mix and every industry's emission intensity at their base-year values. Line (2) additionally lets the industry mix move, so the gap between (1) and (2) measures composition. Line (3) is realised emissions, so the gap between (2) and (3) measures technique. Choi et al. (2023) split that technique effect further, into the part contributed by surviving establishments and the part contributed by entry and exit; the Swedish energy survey does not carry that split, for the reason given in the note to Figure 2.

Figure 2 tracks aggregate manufacturing CO_2 relative to 2006. The black line isolates the scale effect, the blue line adds composition, and the green line is realised emissions, so the residual gap is technique. Scale works against the decline, since output grew over the period. That composition pushes the other way is what Ustyuzhanina (2022) also report for Swedish manufacturing over 2007–2017: the sector moved toward more pollution-intensive goods, and technique had to offset both that and the growth in output.

E Abatement investment

The abatement survey and what it covers

Statistics Sweden runs an annual survey in which firms report investment and current expenditure on environmental protection, in thousands of SEK, split by purpose: clean air, water, waste, and environmental research and development. The survey would answer a question the main text leaves open, which is whether firms that offshore carbon spend less on abating what remains at home.

It cannot, and the reason is the instrument rather than the outcome. The survey draws all manufacturing firms above 250 employees, half of those between 100 and 249, and a fifth of those between 50 and 99, reaching roughly 700 firms a year. On that subsample the shift-share has almost no power. The extensive margin—an indicator for reporting any environmental investment, defined on every surveyed firm-year and conditioning on nothing—has a Kleibergen–Paap statistic of **0.08**. Conditioning on positive investment, which is the obvious worry, is not what binds: the same outcome under the inverse hyperbolic sine of Akerman et al. (2024), which keeps the zeros and so uses the same firm-years, returns a first stage below 10 as well. Every estimate in Table E.1 Panel B carries an interval wide enough to contain a hundredfold increase and a hundredfold decrease.

I therefore report no finding on abatement in either direction. These rows are not evidence that abatement is unchanged: a first stage of 0.08 is uninformative, and the discipline that forbids treating the emissions-in-levels estimate at $F = 8.6$ as a null applies here with far more force. Akerman et al. (2024), using the same registers with a value-weighted exposure measure and a longer window, find that importing raises abatement investment. Nothing here contradicts that, and nothing here supports it.

What the main text does establish about technique rests on different and much stronger evidence. On the largest first stage in the paper, $F = 22.4$, no fuel share responds to the shock and emissions per gigajoule do not move; the domestic-technique margin of Section 7.4 is insignificant at the preferred rung in the pooled sample and in both halves of the split. The argument that the decline is not a technique effect does not depend on the abatement survey and is not weakened by its silence.

Table E.1: What else the shock moves: productivity and abatement

Outcome	N	2SLS	s.e.	KP F
Panel A. Productivity				
Total factor productivity (ln, Wooldridge)	39,804	*0.033*†	(0.102)	9.71
Value added per employee (ln)	22,259	0.217**	(0.087)	13.12
Sales per employee (ln)	22,430	0.180***	(0.065)	13.79
Panel B. Abatement				
Reports any abatement investment (0/1)	3,217	*-2.098*†	(7.675)	0.08
Investment, levels (ln), investors only	621	*-0.108*†	(1.923)	0.68
Investment per VA (ln), investors only	594	*-0.537*†	(1.377)	1.35
Akerman asinh, F series (expenditure)	3,309	*-5.726*†	(11.378)	0.34
asinh, B series (investment)	3,217	*-13.639*†	(50.451)	0.08
Panel C. Reduced form on the shock				
Total factor productivity (ln)	39,804	0.007	(0.023)	–
Value added per employee (ln)	22,259	0.076***	(0.022)	–
Sales per employee (ln)	22,430	0.064***	(0.018)	–
Akerman asinh, F series (expenditure)	3,309	-0.753	(0.718)	–
asinh, B series (investment)	3,217	-0.818	(0.718)	–
Reports any abatement investment (0/1)	3,217	-0.126	(0.103)	–

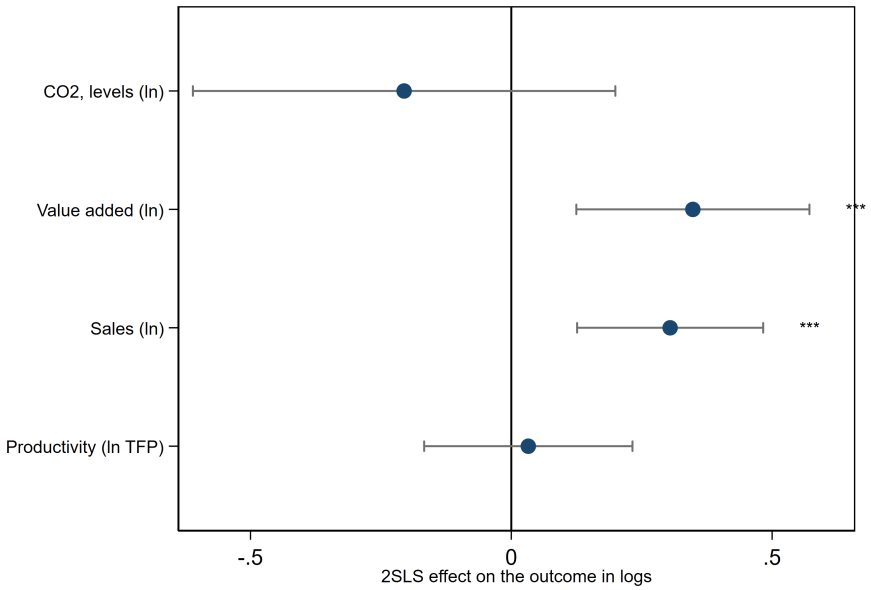
Notes. Each row is a separate 2SLS regression of the stated outcome on log embodied carbon imports, instrumented by the shift-share, with firm, sector–year and municipality–year effects and standard errors clustered by firm. The reported F is the Kleibergen–Paap rk Wald statistic. A dagger marks a row whose first stage is below 10; its stars are suppressed. Panel A reports three measures of productivity because total factor productivity is itself estimated. Panel B runs on the SCB environmental-protection survey, discussed in the text above.

F All outcomes on the preferred specification

Section 6.5 reports the outcomes that bear on the mechanism. Figure F.1 collects all of them estimated on column (3) of Table 2, which shows the ones the text does not discuss and allows the adding-up to be checked: because $\ln(CO_2/VA) = \ln CO_2 - \ln VA$, the coefficient on the headline intensity must equal the coefficient on emissions in levels minus the coefficient on value added.

Two rows carry a first stage below 10 and are drawn accordingly. Emissions in levels sit at $F = 8.6$ and total factor productivity at $F = 9.71$. Neither is a null. The productivity row in particular cannot settle whether offshoring made these firms more efficient: a single estimated TFP measure on a marginal first stage is not enough, and what would settle it is the same elasticity on several measures at once—Wooldridge TFP, value added per employee, value added per hour, and capital per worker—so that a movement in the estimated measure alone could be distinguished from a movement in the firm. That evidence is not assembled here.

Figure F.1: Numerator or denominator?



Notes. The same instrument on four other outcomes. Value added rises 0.348^{***} at $F = 13.4$. CO_2 in levels is -0.205 with a standard error of 0.207 at $F = 8.6$, and productivity is 0.033 at $F = 9.7$; both sit below the conventional threshold and neither is a null.

G The origin-group split of the instrument

The natural way to ask the question is to split the instrument by the origin's environmental policy: if arbitrage drives the result, the shock should bite through lax origins and not through strict ones.

I ran that test. It does not identify. Table G.1 reports the first stage for the pooled instrument and for the same instrument restricted to each origin group. Only the pooled version has a first stage worth the name. Restricting to EPS-lax origins gives an F of 5.6, to EPS-strict origins 0.7, and to the rest of the world 0.4, where the 2SLS coefficient of $+7.6$ is a near-zero denominator rather than an estimate. The reason is collinearity. A firm's exposure to supply shocks from the European Union is nearly the same variable as its exposure to supply shocks from everywhere, because the same baseline product shares weight both. Splitting the shock destroys the identifying variation without delivering six usable instruments. The same happens to every other cut of this instrument, including the split by firm type I come to below: it recovers an average and will not be subdivided.

Table G.1: Splitting the instrument by origin group destroys the first stage

Shock restricted to	First-stage F	2SLS
All origins (pooled)	**18.4**	$\mathbf{-0.504}^{***}$
China / Hong Kong	7.9	-0.075
EPS-lax	5.6	-0.302
European Union	1.4	-0.287
Low income	1.1	-1.155
EPS-strict	0.7	-1.510
Rest of world	0.4	$+7.617$

Notes. Each row re-estimates column (3) of Table 2 with the shift-share built on the stated origin group alone. The reported F is the Kleibergen–Paap rk Wald statistic. **Only the pooled instrument has a first stage.** The rest-of-world coefficient of $+7.6$ divides by a first stage near zero and is not an estimate of anything. EPS classifies an origin as lax if its OECD Environmental Policy Stringency index lies below Sweden's in the same year; Appendix C describes the index.

H Heterogeneity

Two firm characteristics should matter for policy if the response is uneven: a carbon price that already binds on the firm, and a sector where carbon is a large share of costs. Neither turns out to move the estimate, and the way the test is built is what makes that statement worth anything.

Splitting a sample and comparing two confidence intervals tests nothing. I therefore cross the fixed effects with the moderator, so that the interacted model reproduces the split-sample estimates exactly while the *difference* becomes a parameter with a standard error. Table H.1 reports both moderators this way.

Neither difference is distinguishable from zero. Firms in carbon-intensive sectors give -0.575 and firms in the rest of manufacturing -0.657, a gap of 0.082. The data cannot detect whether the response depends on how carbon-intensive the firm's sector is: the difference is 0.082 with a standard error of 0.543.

The EU Emissions Trading System is the more interesting of the two and the weaker. Firms outside the system give -0.677 and firms inside it 0.433. Read naively this says the scheme switches the effect off. I do not read it that way, because the covered cell is small and its interval runs from roughly -1.35 to $+2.22$: it contains the pooled estimate, zero, and effects twice the pooled estimate in the opposite direction. This row is a statement about how few covered firms the sample holds, not a statement about carbon pricing, and I report it so that the absence of a test is on the record.

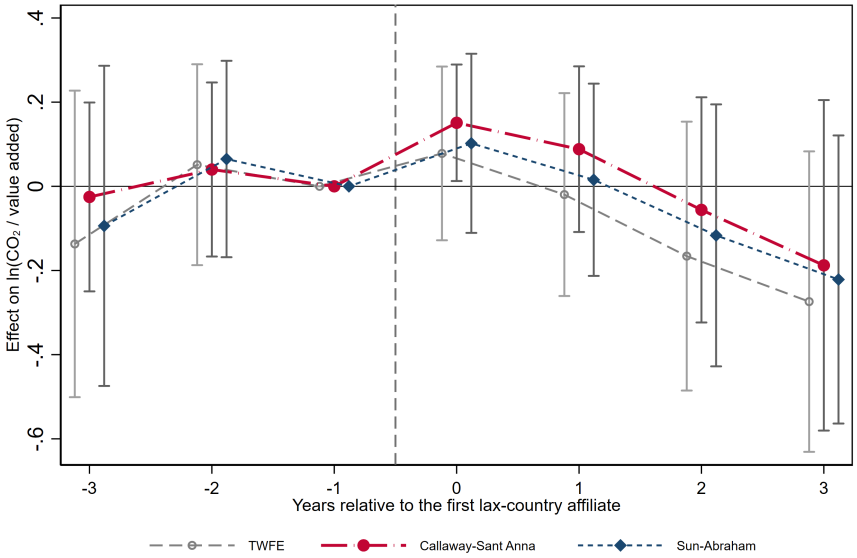
Table H.1: Heterogeneity: sector carbon intensity and the EU ETS

Moderator	Outcome	Low	High	Difference
Sector intensity	CO2 intensity	-0.657	-0.575*	0.082
		(0.421)	(0.343)	(0.543)
EU-ETS	CO2 intensity	-0.677**	0.433	1.110
		(0.294)	(0.910)	(0.956)

Notes. Fixed effects are crossed with the moderator, so the interacted model reproduces the split-sample estimates exactly and the *difference* is a tested parameter rather than a comparison of two confidence intervals. Instrument as in Table 2. **Neither difference is significant.** The ETS cell is small, so that row is a statement about power rather than about carbon pricing. Standard errors clustered by firm. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$.

I Opening a plant abroad

Figure I.1: Opening a plant abroad and the parent's emissions



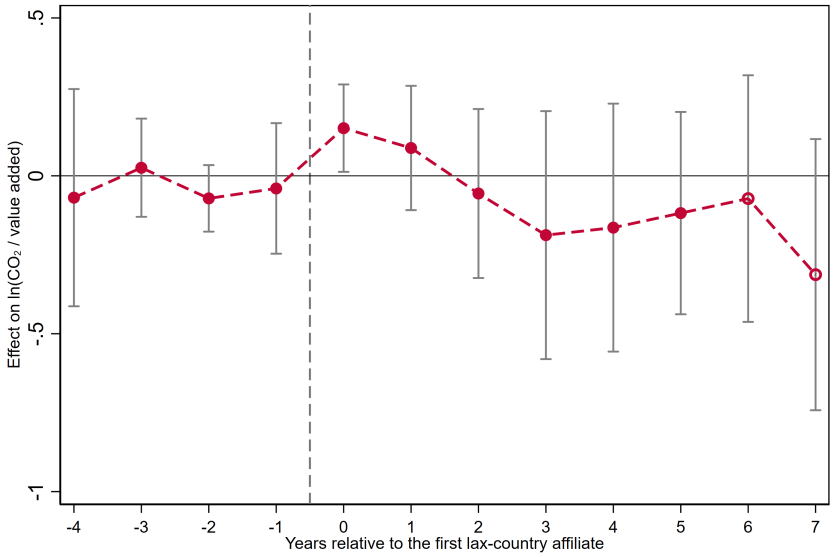
Notes. Callaway–Sant'Anna, Sun–Abraham and TWFE event studies around a firm's first affiliate in a new country, with firms that never open one as the never-treated group. Pre-trend $F = 1.06$, $p = 0.348$. Roughly twelve hundred treated firm-years, so the intervals are wide. Appendix I extends this to every horizon the panel supports and maps where Swedish affiliate employment sits.

Figure I.1 reports the response of a parent's emission intensity around its first foreign affiliate at the horizons the main text uses. Figure I.2 extends it to every horizon the panel supports. The long horizons are not comparable with the short ones: a 2005–2014 window cannot follow a late entrant far forward or an early one far back, so each horizon describes a different set of firms, and the markers resting on two cohorts or fewer are drawn hollow. The aggregates are the objects to read.

Figure I.3 shows where the affiliate employment actually sits. It is concentrated in high-

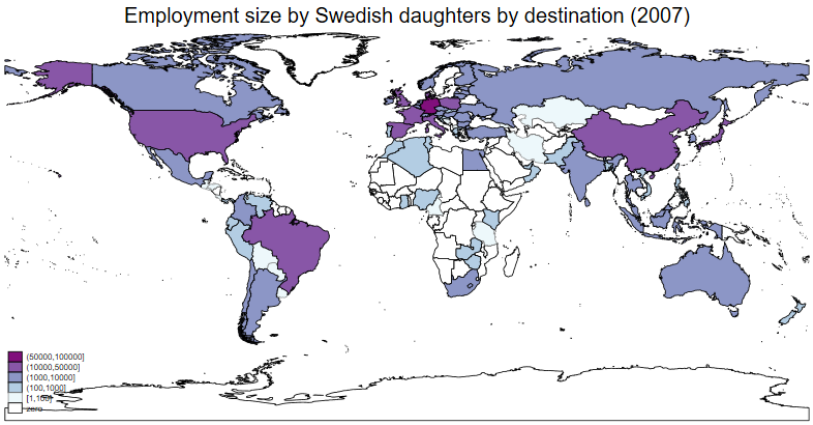
income Europe and North America, not in the destinations the pollution-haven hypothesis points to, which is a descriptive fact about the FDI register, not evidence about emissions, and is the reason the ownership channel is small before any estimation.

Figure I.2: Opening a plant abroad: longer horizons



Notes. Callaway–Sant'Anna only, at every horizon the panel supports. Hollow markers rest on two cohorts or fewer: a 2005–2014 panel cannot observe a late entrant far forward or an early one far back, so the estimate at a long horizon covers a different set of firms from the estimate at zero. Read the aggregates rather than any single horizon.

Figure I.3: Where Swedish affiliates are



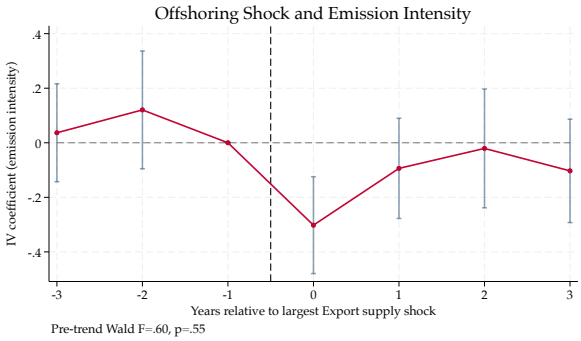
Notes. Employment in foreign affiliates of Swedish manufacturing firms. Employment rather than plants or sales because the missing and zero rate is far lower. *Source:* FDI register, not Swedish firm emissions.

J The reduced form over event time

Section 6.3 traces the instrumented response over event time. That exercise divides by the first-stage coefficient, so a sceptic can object that its shape inherits whatever the first stage is doing. The reduced form does not. Here the event-time dummies are built from the timing of the firm's largest export-supply shock, the outcome is emission intensity, and nothing is instrumented: the figure is a regression of the outcome on the instrument's timing, and its validity does not depend on the strength of the first stage.

Figure J.1 reports it, with the year before the shock normalised to zero. The leads are jointly indistinguishable from zero, with a Wald F of 0.60, so the firms that receive the largest shocks were not already on a declining path. The response begins at the shock and grows, which is the same shape the instrumented version produces. I report this in the appendix instead of the main text because it answers a narrower question than Figure 5 does—it establishes that the timing is not incidental, but it recovers no magnitude—and because it is not an elasticity.

Figure J.1: Reduced form around the largest export-supply shock



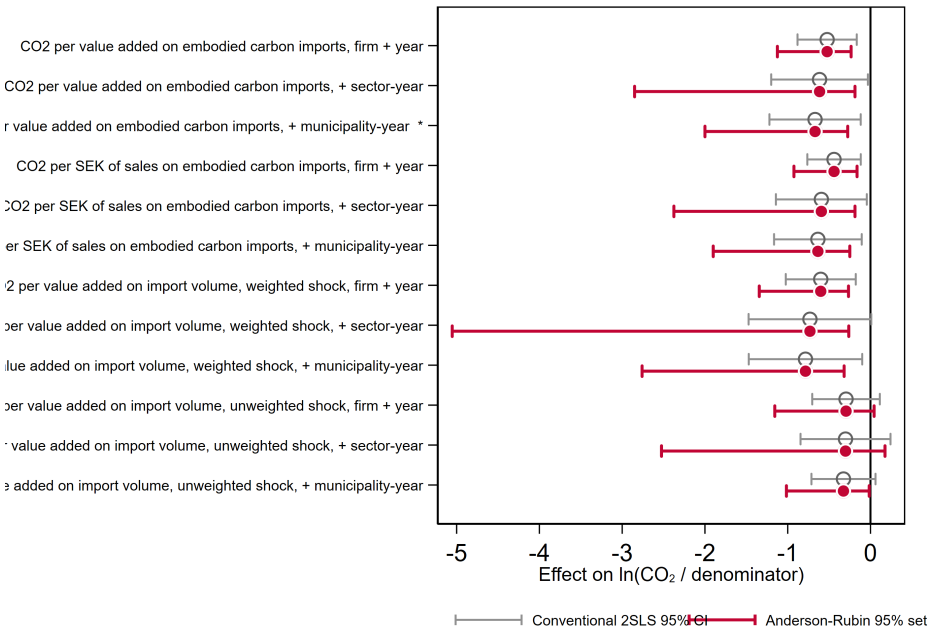
Notes. Emission intensity around the firm's largest shock, $t = -1$ normalised. Nothing is instrumented: the regressors are event-time dummies built from the *shock's* timing, so this is a reduced form. Pre-trend Wald $F = 0.60$, $p = 0.55$.

K Weak-identification-robust inference

The first stage is moderate, not strong, so the conventional two-stage-least-squares interval is not automatically trustworthy: its coverage depends on the first stage being large, and at a Kleibergen–Paap F near sixteen that is an assumption, not a fact. I therefore invert the Anderson–Rubin test over a grid of candidate values of the structural parameter and report the set of values the data cannot reject. Anderson–Rubin coverage is correct whatever the strength of the instrument, including when it is arbitrarily weak.

Figure K.1 reports those sets beside the conventional intervals, at each rung of fixed effects and for both denominators. Every carbon-weighted specification excludes zero, so the headline result does not depend on the conventional approximation, and so does the import-value specification instrumented by the unweighted shock, which is column (6) of Table 2. Column (5) needs no panel of its own: it shares an outcome, an instrument and a sample with column (3), so at $\beta_0 = 0$ the Anderson–Rubin statistic reduces to the same reduced-form regression in both, and the two columns cannot differ in whether they exclude zero. Section 6.1 draws the comparison these estimates support.

Figure K.1: Anderson–Rubin sets against conventional intervals

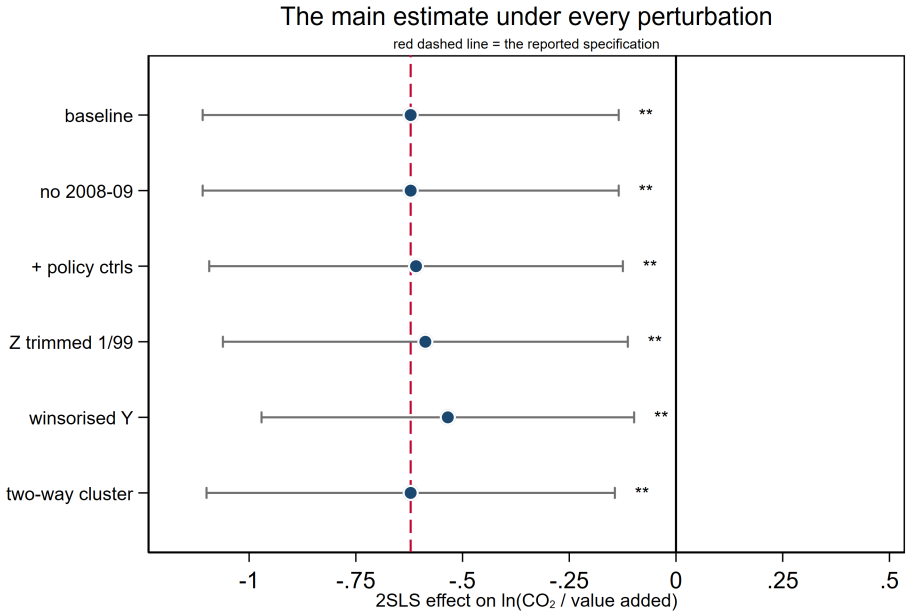


Notes. Each row is one specification, with the conventional 2SLS 95 percent interval above and the Anderson–Rubin 95 percent set below; the marker is the 2SLS point estimate in both. Sets are obtained by inverting the Anderson–Rubin test over a grid of candidate values, which is that test exactly when the model is just-identified. Rows are grouped so that the three fixed-effect rungs of a given specification sit together, and the asterisk marks column (3) of Table 2, the preferred specification. AR coverage does not depend on the strength of the first stage, so a set much wider than the interval above it means that interval was borrowing strength the instrument does not have.

L Specification choices in the shift-share design

The preferred column rests on three kinds of choice: which fixed effects absorb the variation, how the standard errors are clustered, and how emissions are scaled. Each is defensible. None is implied by the design, and until now none had been varied on its own.

Figure L.1: Robustness of the IV estimate

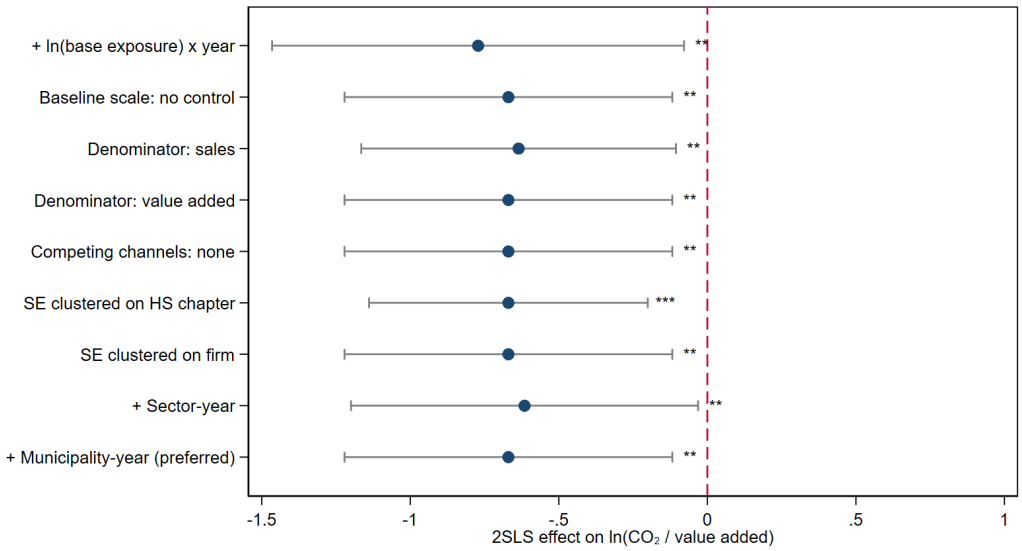


Notes. Baseline; dropping 2008–09; adding policy controls; the instrument trimmed at the first and ninety-ninth percentiles; the outcome winsorised; and two-way clustering on firm and sector–year. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$.

In Figure L.2, I re-estimate column (3) of Table 2 under nine such choices, one at a time. Every interval excludes zero. Absorbing sector–year effects without municipality–year effects gives -0.615. Interacting baseline exposure with year moves the estimate further from zero, not towards it, so the result is not an artefact of firms that started the period heavily exposed. Clustering on the originating HS chapter instead of the firm tightens the interval, which is what a shock common within a product chapter and not within a firm should do. The elasticity survives each choice separately.

The rung with firm effects alone is the exception, and it is omitted from the figure. Its interval runs from roughly -25 to $+8$. Without sector–year effects the instrument has too little usable variation to say anything, so that rung is uninformative about the elasticity, not evidence against it, and plotting it would compress every other row into a sliver.

Figure L.2: The carbon-import elasticity across specification choices

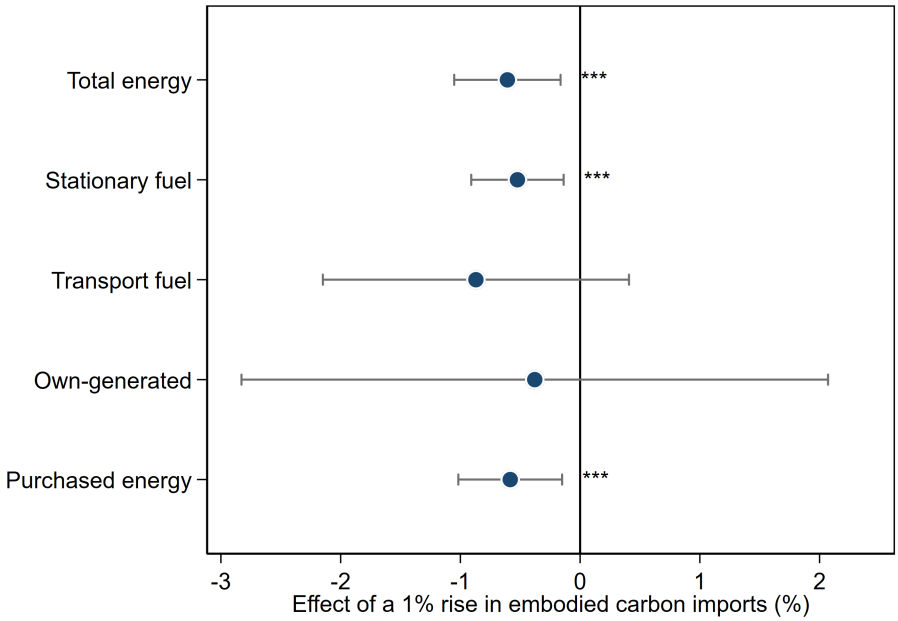


Notes. Each row re-estimates column (3) of Table 2 changing one specification choice and holding the rest fixed. The marker is the two-stage-least-squares point estimate and the horizontal line is its 95 percent interval; stars beside the interval denote significance at the 10, 5 and 1 percent levels. The dashed line marks a zero elasticity. Rows that impose a choice the preferred specification already makes reproduce it exactly, and are kept so that the full set of choices examined is visible rather than only those that move the estimate. The rung with firm effects alone is omitted from the figure: its 95 percent interval runs from about -25 to $+8$, which is uninformative about the elasticity rather than evidence against it, and on a shared axis it compresses every other row.

M The energy identity on quantities

Figure 7 splits the headline elasticity into energy per unit of output and the carbon content of that energy, both as intensities. Figure M.1 runs the same split on the quantities underneath: total energy in gigajoules and total CO_2 in tonnes. Energy is what the SCB survey measures directly and CO_2 is constructed from it by fuel-specific emission factors, so the two rows add exactly and the figure is a check on the construction, not independent evidence.

Figure M.1: Energy and carbon components



Notes. Energy is the primitive the survey measures and CO_2 is constructed from it by fuel-specific factors, so this split is an identity rather than a second estimate.

N Robustness of the decomposition

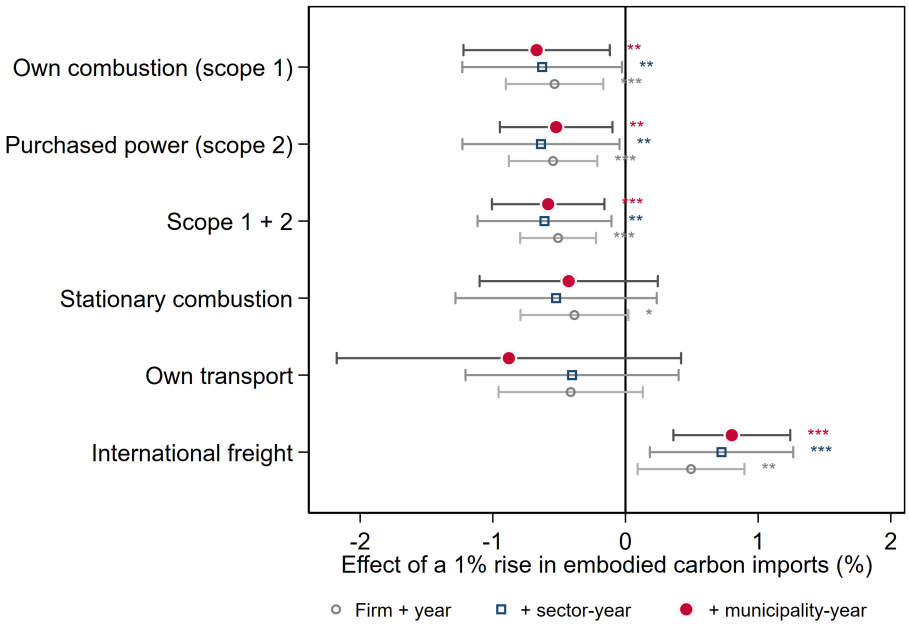
Figure N.1 repeats the component decomposition at each of the three fixed-effect rungs. The purpose is not to find the specification that gives the largest coefficient but to show which conclusions survive all three and which do not, since a result that appears only at the loosest rung is a result about what the looser specification fails to absorb.

O The aggregate decomposition

Swedish manufacturing cut its emission intensity by 94 log points between 2006 and 2014. In Table O.1 I split that change into the part occurring inside continuing firms and the part due to reallocation between firms together with entry and exit, and in Figure O.1 I set the within-firm term against the change the estimated elasticity attributes to offshoring.

The two panels of the table measure the same decline on different scales and do not agree about it. In log points, continuing firms account for 91.5 percent of the fall. In levels they account for 48 percent, and net entry for 68 per cent, with reallocation between survivors working the other way. A level is dominated by the largest and dirtiest plants and moves sharply when a few of them close, which is what drives the two apart. I report the log decomposition in the body

Figure N.1: Components across fixed-effect rungs



Notes. The decomposition of Figure 6 re-estimated at successively richer fixed effects.

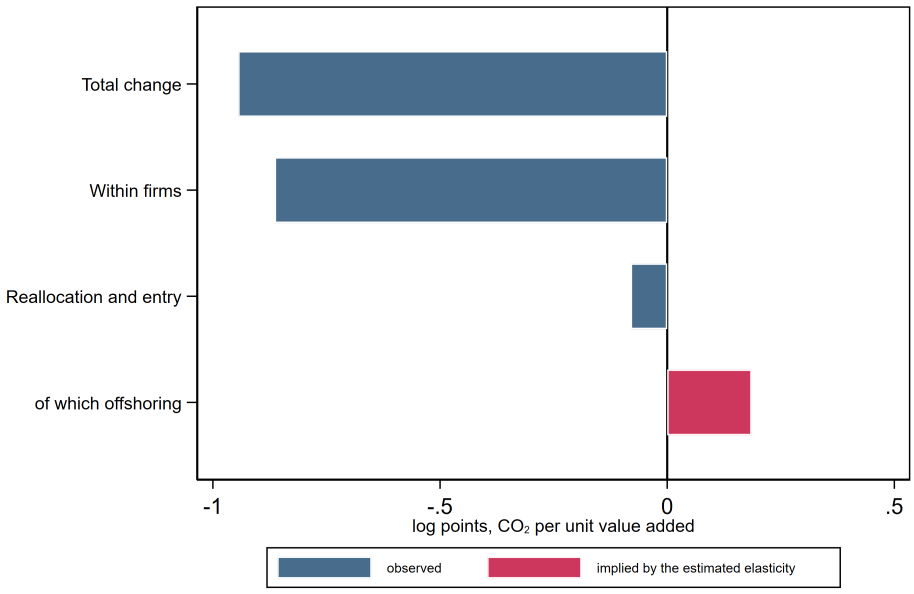
because the quantity attributed to offshoring is itself a log-point quantity: it is formed as the value-added-weighted sum across firms of $\hat{\beta}$ times each firm's change in log embodied carbon imports. It is commensurable with Panel B and not with Panel A.

The attributed quantity carries the opposite sign because of the regressor rather than the elasticity. Among firms observed in both the first and the last year of the window, the value of what they imported rose by 21 log points while the carbon embodied in it fell by 11, so carbon per krona of imports fell by 32. Firms bought more from abroad, and what they bought grew cleaner faster than the volumes grew.

Nothing in the construction forced that answer. Had the carbon embodied in these firms' imports grown with the value of those imports, the same elasticity over the same window would have attributed a substantial part of the within-firm decline to offshoring. The attribution runs against the decline because carbon per krona of imports fell, not because the design cannot detect the channel: column (3) of Table 2 estimates that channel on an interval excluding zero. The magnitude here is a fact about what Swedish firms bought over this decade, not a property of the method.

That attributed quantity asks the estimated elasticity to do something the research design does not identify. I estimate $\hat{\beta}$ within sector–year and municipality–year cells, so it compares firms facing common aggregate conditions and sets aside precisely the movement over time

Figure O.1: The decline in emission intensity, and what offshoring accounts for



Notes. Log points of CO_2 per unit of value added, Swedish manufacturing, 2006–2014. The first three bars decompose the observed change into the part occurring inside continuing firms and the part due to reallocation between firms together with entry and exit; they sum to the total by construction. The fourth bar is not observed. It is the change the estimated elasticity attributes to offshoring, formed as the value-added-weighted sum across firms of $\hat{\beta}$ times each firm's change in log embodied carbon imports, with $\hat{\beta}$ taken from column (3) of Table 2. It carries the opposite sign to the decline because embodied carbon imports fell over this window. The text states what carrying the elasticity to the aggregate requires.

that the decomposition is about. Carrying it to the aggregate requires that the relative effect survives aggregation: that one firm's offshoring does not displace another firm's emissions, and that input and output prices do not adjust. The exercise is therefore accounting under a stated assumption rather than a counterfactual. The conclusion in the body survives the qualification, because the attributed effect does not merely fall short of the decline. It carries the opposite sign, and the assumption would have to fail both severely and in one particular direction to reverse that. The required failure is specific. Displacement would have to exceed the attributed effect in magnitude and reverse its direction: about one and a half times that effect to account for a tenth of the within-firm decline, and between five and six times to account for all of it.[8]

[8]From Table O.1, with log changes scaled by one hundred: the within-firm decline is 86.4 log points and offshoring is attributed 18.6 in the opposite direction. A tenth of the decline is 8.6, so displacement must deliver $8.6 + 18.6 = 27.2$ log points, which is 1.5 times the attributed effect; the whole decline requires 105.0, which is 5.6 times.

Table O.1: The decline in emission intensity, 2006–2014

Panel A. Levels: kg CO_2 per SEK of value added		
	Change	Share of total
Total, 2006–2014	-0.032473	100.0
within firms	-0.015464	47.6
between firms	0.005023	-15.5
net entry	-0.022032	67.8
Panel B. Log points		
	Change	Share of total
Total, 2006–2014	-0.9444	100.0
within firms	-0.8640	91.5
reallocation and entry	-0.0804	8.5
Implied by the estimated elasticity, same firms		
within term, these firms	-0.8640	
attributed to offshoring	0.1859	

The implied effect and the within-firm decline have **opposite signs**: embodied carbon imports fell over this window, so a negative elasticity implies offshoring raised intensity. A share of the decline is therefore not defined and none is reported.

Notes. Panel A measures the change in kilograms of CO_2 per krona of value added; Panel B measures it in log changes, which the text quotes as log points after multiplying by one hundred, so the -0.9444 in the first row of Panel B is the 94-log-point fall discussed above. Within-firm, between-firm and net-entry terms sum to the total by construction in each panel. The final block applies the elasticity from column (3) of Table 2 to each firm's observed change in log embodied carbon imports and aggregates with base-year value-added weights.

P Constructing the embodied-carbon measure

Intensities

EXIOBASE 3 reports, for each of 49 regions and about 200 products, gross output x and two emission vectors: direct emissions and emissions embodied in the full upstream chain, the second obtained from the Leontief inverse $\mathbf{S}(\mathbf{I}-\mathbf{A})^{-1}$. I take the product-by-product release for 2004–2014 and use the embodied measure throughout; the direct measure is built alongside it and used only to check that the two carry different information. Their correlation across product-region-years is well below one, so the upstream component is not redundant.

EXIOBASE reports kilograms of CO_2 per million euro of output. I rescale to tonnes per thousand euro by a factor of 10^{-6} and hold the currency conversion at a constant rate, so that exchange-rate movement cannot enter the intensity. No deflator is applied: intensities are in nominal euro of the same year as the trade value they multiply, and firm and year effects absorb a common annual price index, so this matters for descriptive levels and not for the estimates.

From EXIOBASE products to HS lines

Swedish customs records an eight-digit CN code; EXIOBASE records a product. A concordance maps HS6 codes to EXIOBASE products, and it is not one-to-one. Where an HS6 code maps to a single product the intensity is merged directly. Where it maps to several, I take the output-weighted average across them, so a product that is a small part of the HS line's output counts for little. Two further rules handle gaps, and both follow Leisner et al. (2026):

- **Zeros.** If every observation of a product-region is zero the intensity is zero, because the product genuinely records no emissions. If only some are zero they are treated as reporting gaps and replaced by the mean of the non-zeros. Otherwise the plain mean applies.
- **Missing years.** Where a product-region has no intensity for a given year, or the entry is zero, the across-year mean for that product-region is used instead.

HS codes are zero-padded to six digits before matching. Read as numbers they lose the leading zero on chapters 01 to 09, which covers live animals, meat, dairy, fish and vegetables, and those lines can then never match.

Partner countries to EXIOBASE regions

EXIOBASE has 44 named countries and five rest-of-world aggregates, covering Asia and the Pacific, Europe, Africa, America and the Middle East. The Swedish trade register has roughly two hundred partners. Every partner outside the 44 collapses into one of the five aggregates and loses its own intensity.

This is a cost and not a detail. The instrument is identified partly by which country a firm buys from, so a partner pushed into an aggregate shares an intensity with a continent. The bridge script reports how much Swedish import value ends in an aggregate, and that share is the right measure of how much the collapse can matter.

Aggregation to the firm, and the Swedish counterfactual

For firm i in year t, summing over import lines indexed by product p and origin c,

$$C_{it}^{for} = \sum_{p,c} M_{ipct}\, e_{pct}, \qquad C_{it}^{swe} = \sum_{p,c} M_{ipct}\, e_{pt}^{\mathrm{SE}}, \tag{11}$$

where M is the value of the line and e the embodied intensity. The two differ only in whose technology prices the basket: C^{for} at the origin's intensity, C^{swe} at Sweden's for the same product.

C^{swe} is the domestic counterfactual used in equation (8) and the denominator of the origin-dirtiness margin.

I also build C^{swe0}, which fixes Swedish intensities at their base-year values. Because e^{SE}_{pt} moves over time, $\ln(C^{swe}/M)$ responds both to what the firm buys and to how Swedish technology changes; holding the Swedish vector fixed isolates the first.

Swedish intensities are winsorised before use, at cut points the log reports.

Units

As built in (11), C^{for} multiplies nominal SEK by kilograms per million euro. That hybrid is fine for ratios, where the value weights cancel, and it is not a mass, so freight cannot be added to it. Converting the value leg gives

$$\text{kg CO}_2 \;=\; \frac{M_{ipct}}{10^6 \times \text{SEK per EUR}} \;\times\; e_{pct},$$

at the Eurostat annual average rate. A same-year nominal conversion is the correct one, since the trade value is nominal SEK and EXIOBASE output is nominal euro of the same year. A purchasing-power rate would not be.

What the measure does not do

Intensities are industry-by-country and not firm-by-product, because producer-level intensities for source countries do not exist. Two German suppliers with different technologies are therefore assigned the same intensity, which is measurement error in the regressor and the reason the estimates are instrumented.

The upstream chain in $\mathbf{S}(\mathbf{I}-\mathbf{A})^{-1}$ also includes Swedish inputs to foreign production. Those emissions sit inside C^{for} and inside Swedish domestic emissions at the same time, so summing the two counts them twice. Tracing and removing such terms is the subject of Koopman et al. (2014) and, in the form used for applied work, Los et al. (2016). Nothing in the estimates turns on it, since the elasticity is identified from instrumented variation in C^{for} and not from its level. The accounting in Section 7 does add the two, and there the sum is an upper bound.

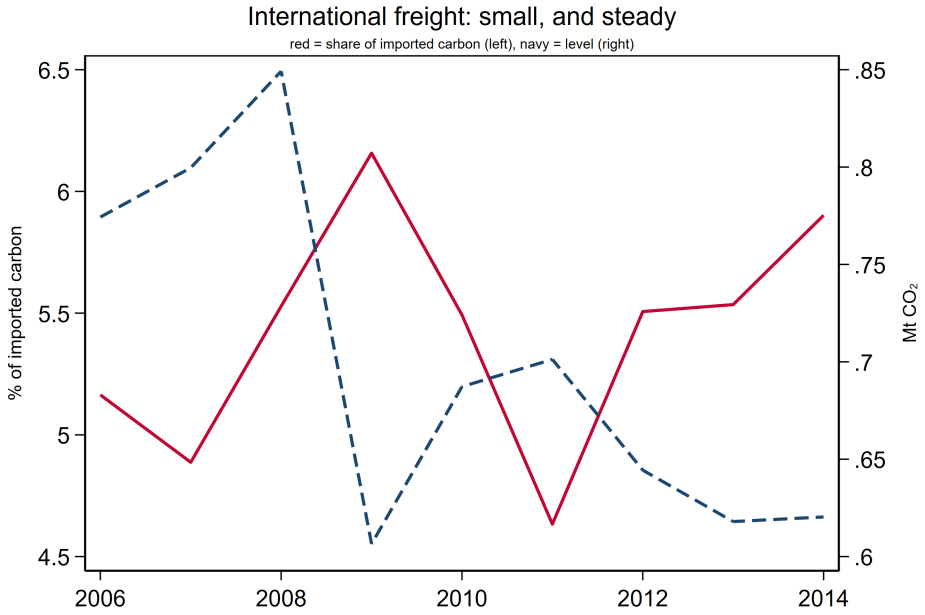
Q Freight-emission construction

I construct international freight emissions as $E^{\mathrm{FR}}_{it} = \sum_c M_{ict}\, d_c\, \varphi_{m(c)}$, where M_{ict} is the weight (in tonnes) of firm i's imports from country c, d_c is the CEPII population-weighted Sweden–c

distance, and φ_m is the CO_2 emission factor (grams per tonne-kilometre) of transport mode m. Central modal factors follow Cristea et al. (2013) and the GLEC framework: sea 15, rail 22, road 100, and air 600 g CO_2 per tonne-km. Mode is assigned by geography—road for contiguous or short-haul intra-EU routes, sea otherwise, with air reserved for high value-to-weight flows. Within this construction, distance varies across origins by more than an order of magnitude and the modal factors span a factor of forty, so mode assignment matters at least as much as distance for E^{FR}. Whether the estimated freight response survives a sweep over the plausible ranges (sea 8–40, road 60–160) and over alternative mode assignments is not established here, and I therefore make no robustness claim for it.

Figure Q.1 reports the resulting series, in levels and as a share of the firms' own reported emissions.

Figure Q.1: International freight emissions: level and share

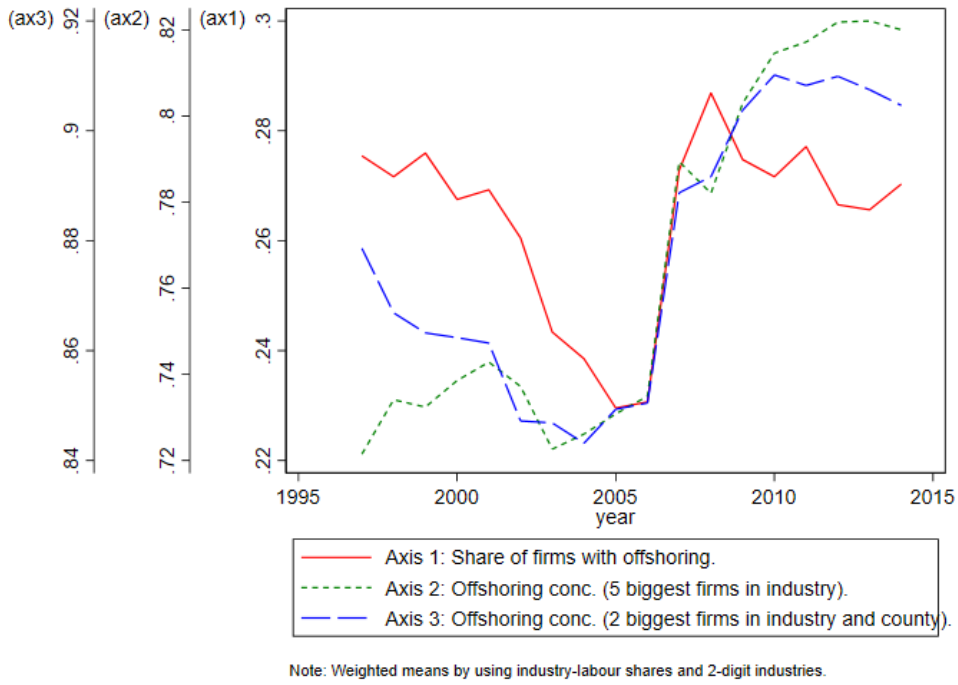


Notes. The dashed line is total international freight CO_2 carrying Swedish manufacturing imports, in megatonnes, against the right axis. The solid line is the same quantity as a percentage of the carbon embodied in those imports, against the left axis. The level falls across the window while the share rises, because embodied carbon imports fell faster than the freight carrying them. Construction is as given above; mode is inferred from geography rather than observed and import weights are net mass, so both series are lower bounds.

R Concentration of carbon offshoring

Section 4 describes how concentrated carbon offshoring is across firms within an industry. Figure R.1 reports the three series that description rests on.

Figure R.1: Concentration of carbon offshoring within 2-digit industries, 1997–2015



Notes. This figure provides insights into the pollution offshoring patterns at the sector level by illustrating the concentration of offshoring within 2-digit industries. The concentration is measured in terms of the offshoring share represented by: (i) the offshoring activities of all firms engaged in offshoring within each industry; (ii) the total offshoring conducted by the five largest offshoring firms within each industry; and (iii) the total offshoring conducted by the two largest firms within each industry and county. Weighted means, using industry–labour shares and 2-digit industries. The three series are plotted against three separate vertical scales, shown at the left, so their levels are not directly comparable across series.